

生产网络中的金融摩擦与货币政策的传导

2026-08-12 · ECB · 发布于 2026-08-06

工作论文系列

JEL: C32, C67, E31, E32, E52

在本模型中，我们从理论上展示了上游成本渠道与下游需求渠道如何在生产网络内部产生并相互作用。

实证结果表明，货币紧缩对产出和价格具有显著的收缩效应。尽管在杠杆率较高的部门中，价格响应略有减弱，但网络维度放大了传导效应，且在数量上更为重要：在峰值时，源于间接金融摩擦——通过生产网络中该部门贸易伙伴的资产负债表发挥作用——的放大效应，约为部门自身杠杆直接效应的三倍。

相关文献。本文处于三支文献的交汇点：货币政策通过生产网络传导的实证分析、金融摩擦在该传导中的作用，以及网络经济中冲击传播的更广泛理论。

转向第二个研究脉络，金融摩擦自Bernanke和Gertler（1995）及Bernanke等人（1999）以来，一直是货币政策传导分析的核心。近期，企业层面的研究将注意力转向该传导过程中的异质性。Ottonello和Winberry（2020）表明，面临财务约束的企业在货币政策冲击后，其投资减少幅度显著大于未受约束的企业，并在异质性企业新凯恩斯模型中对此发现进行了合理解释。Jeenas（2023）强调企业流动性管理作为补充边际的作用，发现持有流动资产较少的企业在加息后投资削减更为剧烈，这一效应独立于杠杆率或企业规模。我们在此文献基础上，超越单个企业视角，展示金融摩擦通过供应商-客户关系进行传播：一个部门的产出和价格对货币政策的响应，不仅取决于其自身资产负债表，还取决于其贸易伙伴的资产负债表状况。我们的发现也与Adelino等人（2023）的研究相关联，他们记录了贸易信贷如何塑造非常规货币政策的传导。我们将该视角拓展至常规利率政策，并利用生产网络结构分别识别需求渠道和成本渠道。

全球金融危机，而Su（2024）证明，投入产出关联会放大部分层面的金融扭曲，从而产生总体全要素生产率损失。我们的模型在Bigio和La'O（2020）的基础上进行了扩展，引入了跨部门的异质性货币政策传导，该传导由部门特定的杠杆率决定，这使我们能够推导出直接、上游和下游金融摩擦渠道的闭式分解，该分解直接映射到我们的实证设定上。

投入产出分析研究（FIGARO）数据库。部门划分大致对应于NACE Rev. 2分类的两位代码层级。¹ 这些表格以年度频率提供，覆盖2010年至2022年。为覆盖整个研究期间，我们将2010年之前的数值设定为与2010年观测值相等，而2022年之后的数值则设定为其2022年的水平。我们之所以这样做，是因为投入产出关联随时间变化非常缓慢。为说明我们从投入产出矩阵中提取的关键指标，表1给出了一个仅包含两个国家（A和B）及两个行业（1和2）的投入产出矩阵示意图。

Table 1: Stylized multi-country input-output table

	A_1	A_2	B_1	B_2	$Final\ consumption$	$Total$
A_1	$x_{11}^{A,A}$	$x_{12}^{A,A}$	$x_{11}^{A,B}$	$x_{12}^{A,B}$	c_1^A	$y_1^A = \sum_{k,d} x_{1k}^{A,d} + c_1^A$
A_2	$x_{21}^{A,A}$	$x_{22}^{A,A}$	$x_{21}^{A,B}$	$x_{22}^{A,B}$	c_2^A	$y_2^A = \sum_{k,d} x_{2k}^{A,d} + c_2^A$
B_1	$x_{11}^{B,A}$	$x_{12}^{B,A}$	$x_{11}^{B,B}$	$x_{12}^{B,B}$	c_1^B	$y_1^B = \sum_{k,d} x_{1k}^{B,d} + c_1^B$
B_2	$x_{21}^{B,A}$	$x_{22}^{B,A}$	$x_{21}^{B,B}$	$x_{22}^{B,B}$	c_2^B	$y_2^B = \sum_{k,d} x_{2k}^{B,d} + c_2^B$
$Labor$	l_1^A	l_2^A	l_1^B	l_2^B		$l = \sum_{k,d} l_k^d$
$Value\ added$	v_1^A	v_2^A	v_1^B	v_2^B		$v = \sum_{k,d} v_k^d$
$Taxes$	t_1^A	t_2^A	t_1^B	t_2^B		$t = \sum_{k,d} t_s^d$
$Total$	y_1^A	y_2^A	y_1^B	y_2^B		$y = \sum_{k,d} y_s^d$

在此记号中， $x_{jk}^{A,B}$ 表示国家A中部门j生产并销售给国家B中部门k的商品流量， c_1^A 表示国家A中部门1的产出用于最终消费的数量， l_1^A 为所使用的劳动量， v_1^A 代表增加值， t_1^A 为扣除补贴后的税收。给定行j中各元素之和等于部门j生产的总产出（用于中间消费和最终消费）。给定列k中各元素之和给出部门k生产中所使用的投入总成本（包括劳动、税收和增加值）。根据构造，总产出等于投入总成本。

矩阵 Ω 可用于捕捉冲击的直接（一阶）效应，例如某一特定部门产出需求的变化对其直接供应商生产的影响。⁴ 然而，它并未考虑沿供应链后续各轮生产调整所产生的间接（高阶）效应。为将这些高阶效应纳入考量，我们遵循Acemoglu等人（2016）的方法，将列昂惕夫逆矩阵定义为：

X

典型事实

按NACE一级分类的行业敞口，欧元区

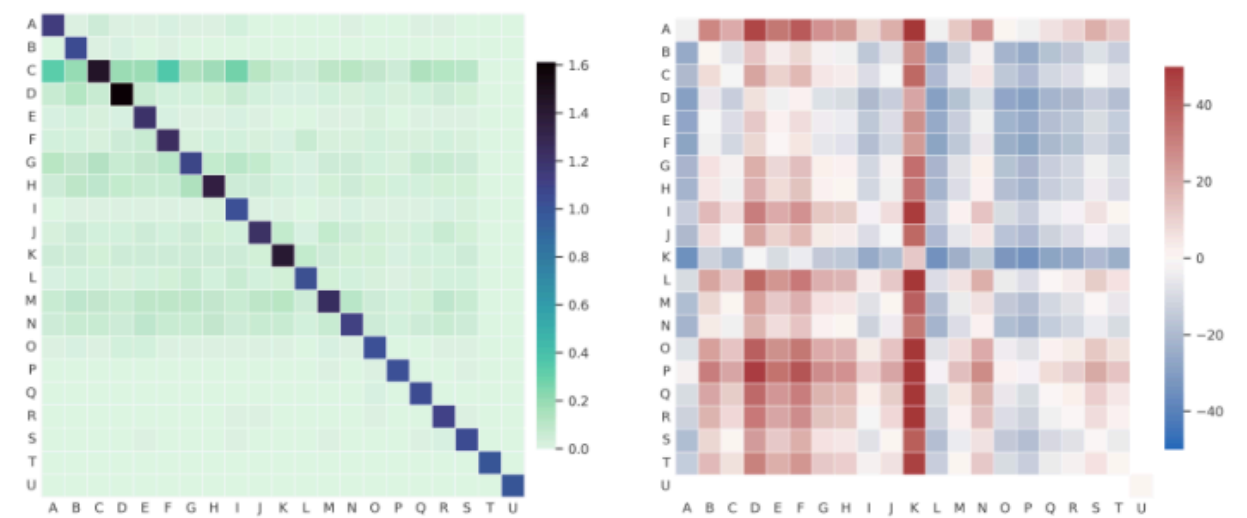


图1：投入产出关联与杠杆引发的上下游重新加权

(a) 列昂惕夫逆矩阵

左下象限汇集了双侧敞口均较低、处于网络边缘的部门，包括房地产（L）、教育（P）、健康（Q）、公共管理（O）、其他服务业（S）和艺术（R）。最后，行政/支持服务（N）位于右下象限，主要通过其供应商暴露于风险。

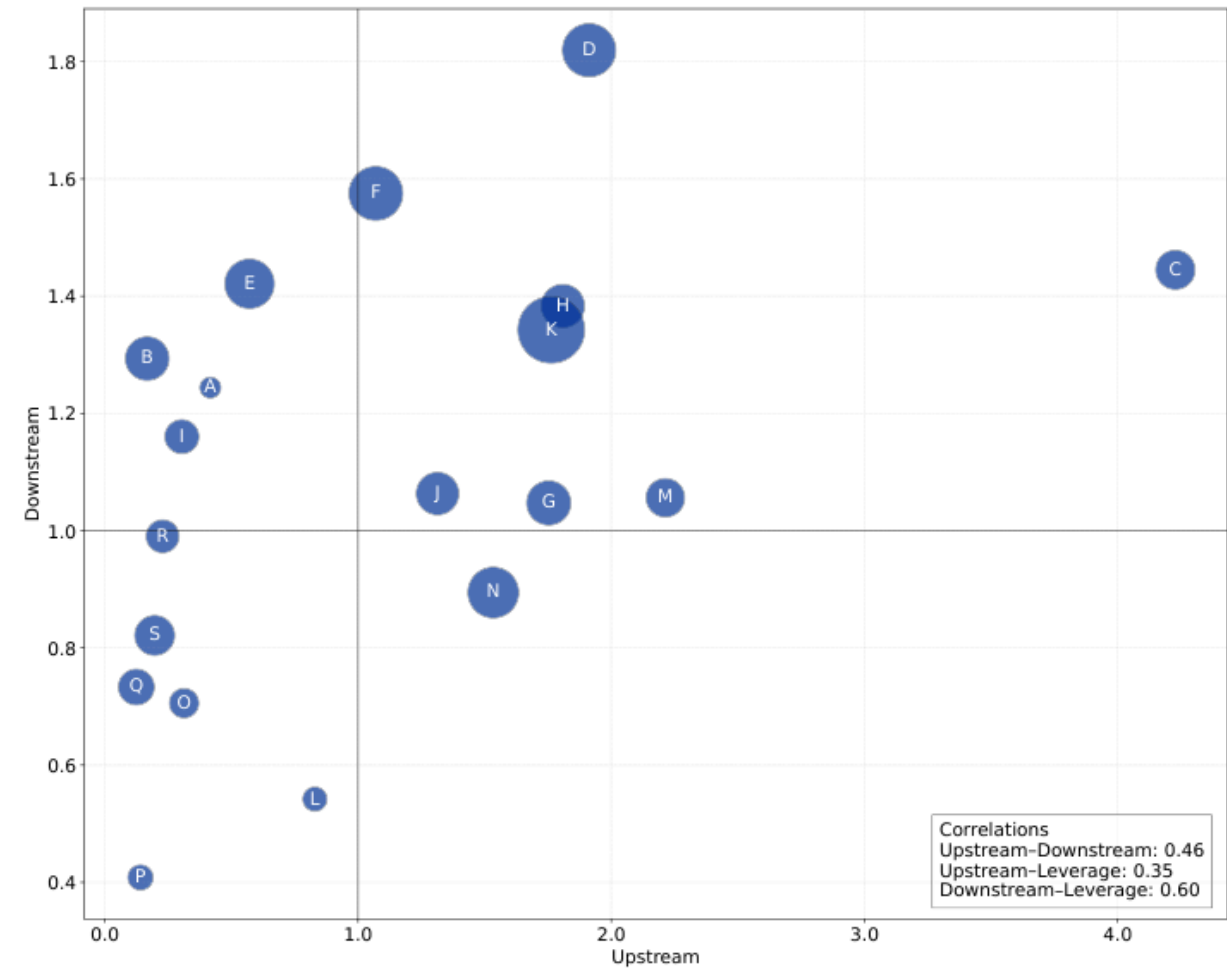


图2：欧元区部门杠杆与生产网络金融敞口

D 1.8 1.6

欧盟统计局短期商业统计（STS）数据集，辅以标准STS覆盖不完整的特定行业补充数据。所有序列构建的完整细节，包括行业调整、季节性调整程序和数据清洗步骤，见附录A。所有变量的汇总统计见表A.3。

Gertler, 1995)。

最后，矩阵 (H_t) 包含我们交互项中与货币政策冲击 (s_t) 无关的其余单一元素，这些元素在我们的分析中并非首要关注对象。矩阵 (K_t) 收集了因变量 $(\Delta y_{j,t})$ 的一阶差分滞后项及冲击变量。矩阵 (X_t) 包含一组宏观金融控制变量，包括欧元区三个月期OIS利率、GDP加权的10年期

欧元区综合主权债券收益率、欧元兑美元汇率，以及欧元区系统性压力综合指标（CISS）的对数水平、IMF大宗商品价格指数、欧元区调和消费者物价指数（HICP）和欧元区失业率，此外还包括部门就业，我们将其作为额外的部门特定宏观经济控制变量。 θ_m 表示月份固定效应， θ^{13} 我们通过添加一个前瞻性虚拟变量 κ_{t+h} 来控制新冠疫情的影响，该变量与因变量在同一响应区间进入模型。 θ^{14}

发现1：面对货币政策紧缩冲击，价格与产出均出现下降，而金融摩擦在中期内放大了这一紧缩效应。

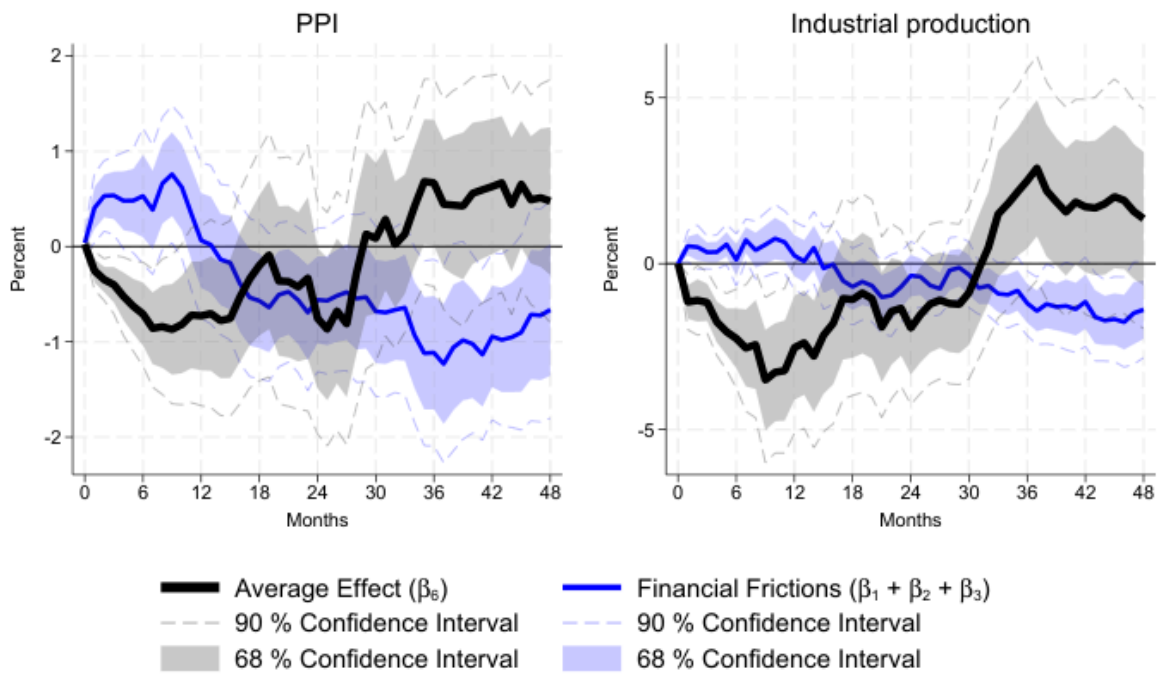


图3：对货币政策紧缩冲击的脉冲响应

PPI

摩擦效应

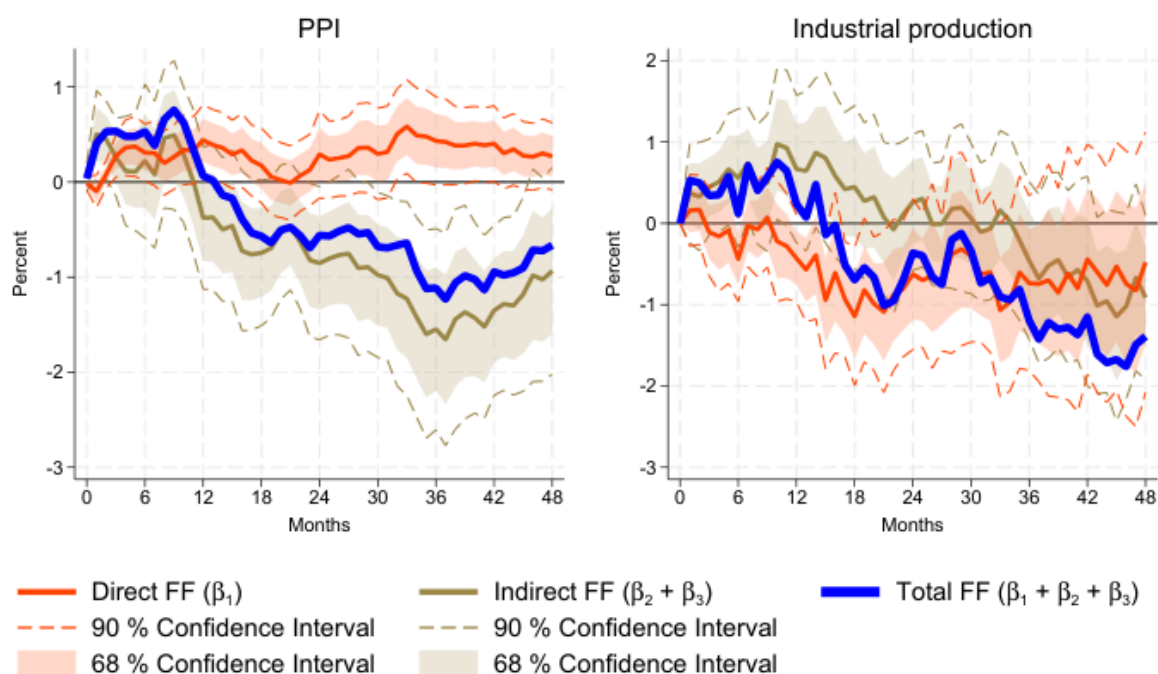


图4：对货币政策紧缩冲击的脉冲响应——直接与间接金融对比

PPI

图5右面板)，而下游金融摩擦则加剧了额外的下降（图5右面板中的粉色线）。综合来看，这些结果表明，金融条件的收紧降低了下游客户对部门i所生产中间品的需求（与部门特定的“需求渠道”一致），同时增强了上游供应商提高价格以缓解金融摩擦的动机（与部门特定的“成本渠道”一致）。

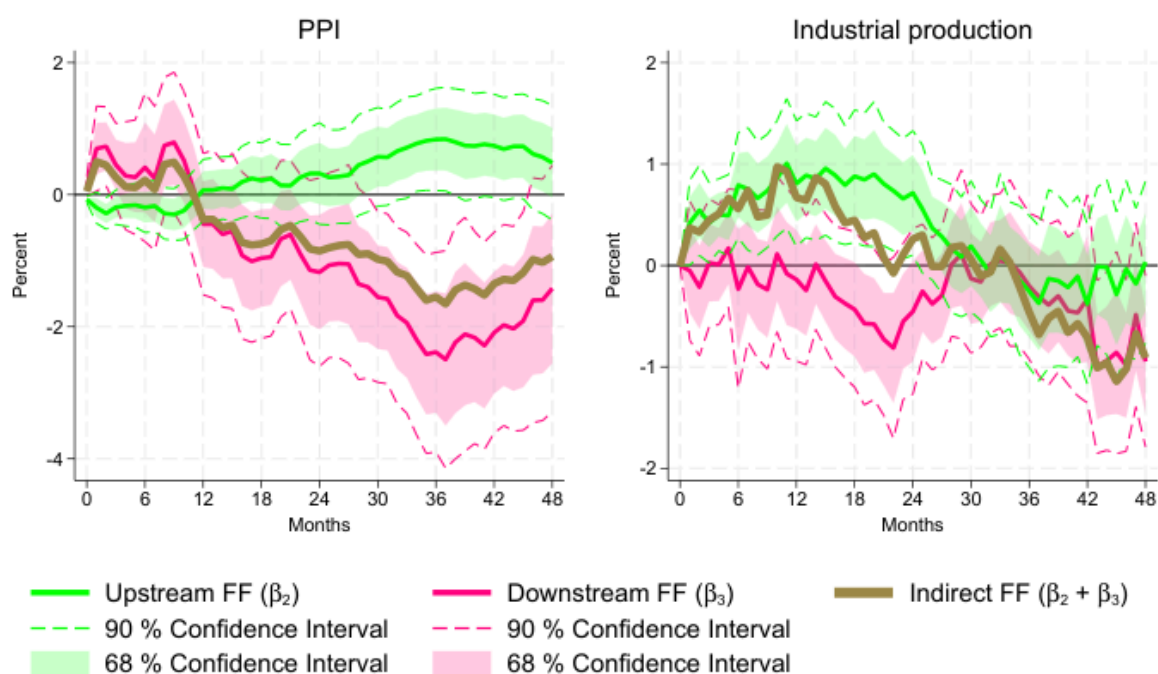


图5：货币政策紧缩冲击的脉冲响应——上游与下游对比

PPI

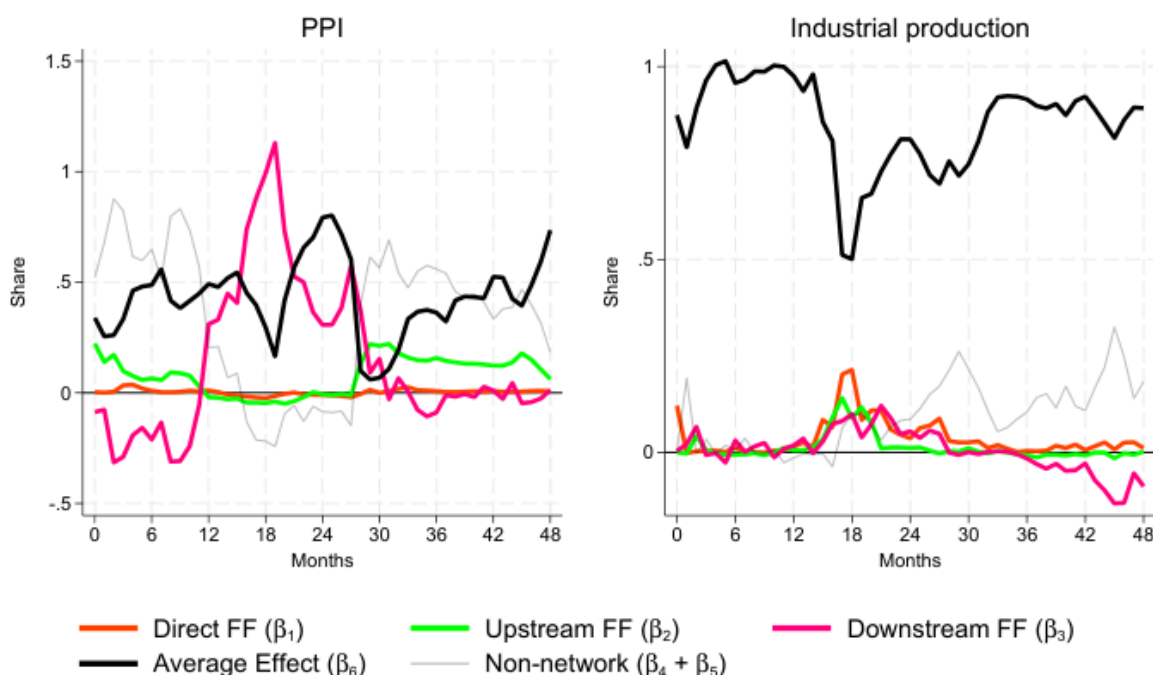


图6：货币政策渠道的方差分解

PPI

对该参数的解释，参考了近期实证与理论研究成果，这些研究表明，杠杆率较高的企业在货币紧缩后，其信用利差上升幅度更为显著。Anderson和Cesa-Bianchi（2024）指出，这一模式可以在一个包含违约风险、面临财务约束的金融中介以及分割金融市场的异质性企业模型中得到合理解释，凸显了中介资产负债表约束在企业层面传导中的核心作用。在该环境中，杠杆率作为企业对外部融资溢价中介驱动波动的充分统计量。我们采纳相同的解释：数据中的部门杠杆率代表对融资成本变化的暴露程度，并在模型中通过部门特定的营运资本摩擦来体现这种暴露。货币紧缩提高了可变生产投入的融资成本，其增幅与该部门的摩擦参数成比例。这种简化形式使我们能够忽略违约和金融中介的详细建模，同时保留杠杆放大货币政策传导的关键机制。

欧拉方程决定了名义支出：货币紧缩通过共同的名义规模效应压低当前总体价格水平。部门层面的响应仍可分解为相同的直接、上游和下游金融摩擦渠道，以及一个在实证设定中被国家-时间固定效应吸收的共同总体项。

企业必须在产生销售收入之前预先借款以覆盖可变生产成本。有效借贷成本随名义利率 i 上升而上升，且在不同部门间存在异质性：参数 $\phi_j \geq 0$ 刻画了部门 j 的有效借贷成本对政策利率变化的敏感度，反映了政策利率向该部门企业所面临贷款利率的传导。这一楔子将总生产成本乘以因子 $1 + i\phi_j$ ，使得企业利润为

投入产出结构与向量记号

价格-利率弹性分解

第三步：求解计价单位条件中的 $\hat{e}$ 。由于 p_c 在所有时点均等于 1，在 $\sigma = 1$ 处对 CES 价格指数求微分可得 $v_T \hat{p} = 0$ 。将此约束施加于微分价格系统，可得

上游金融摩擦

起点是精确分解式24，可写为：

在Cobb-Douglas基准情形下，无需额外的部门层面代理变量来解释这一渠道。

– 在所有模拟经济体中均如此。共同的收入渠道显示出混合的符号，这与理论上的模糊性及其被时间固定效应吸收的情况一致（附录F中的表F.2）。第四，基于方程24中价格-利率弹性完全模型一致性分解的方差分解表明，下游客户渠道在价格变动中占比最大，如附录表F.3右列所示。这与图6中报告的下游金融摩擦效应在生产价格方差中的较大份额一致。在表F.3中，行业内下游客户效应（iii）在变动中占比最大，但部分被供应商客户效应（iv）的负方差份额所抵消。虽然如第4.4节所述，我们无法在实证度量中直接区分这些渠道，但图6中下游金融摩擦效应随时间的符号变化，总体上与理论设定中这些抵消效应一致。

上游与下游摩擦通过截然不同且相互对立的渠道发挥作用。下游行业减少对中间投入品的需求，加剧了价格与活动的下滑。上游行业则通过提价来稳定现金流，部分抵消了通缩压力。下游需求渠道占据主导地位，因此，在货币紧缩之后，网络层面杠杆率的上升会导致价格和产出出现更强烈且更持久的下降。

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A Dataset construction

Euro area and country level data

We collect a set of standard macroeconomic variables at both the euro area and country levels from Eurostat and the IMF, which serve as controls in the empirical analysis. To identify common euro area monetary policy shocks, we exploit high-

frequency movements in the three-month overnight index swap (OIS) rate within a narrow window (approximately 135 minutes) surrounding the publication of the press release and the subsequent press conference following ECB Governing Council meetings. These data are drawn from the euro area extended monetary policy event-study database constructed by Altavilla et al. (2025). To disentangle monetary policy shocks from information shocks, we implement the “poor man’s sign restriction” approach proposed by Jarociński and Karadi (2020). Specifically, we identify monetary policy shocks as high-frequency changes in the three-month OIS rate within the event window that are associated with contemporaneous movements in stock prices of the opposite sign.²⁶

Country-sector level data

We collect data on prices, industrial production, turnover, employment, hours worked, and wages from the “Short-Term Business Statistics” (STS) dataset of Eurostat.²⁷ As no information on price indices for the trade sector (NACE codes G00, G45, G46, G47) are reported in the STS, we compute them by dividing nominal turnover by real turnover, consistently with the methodology employed by Eurostat.²⁸ In addition, we proxy the producer price index (PPI) for the construction sector (NACE code F00) with the PPI for residential buildings construction.²⁹ Moreover, as STS do not cover the agriculture sector (A01), we complement PPI data for this sector with quarterly information from the “Price Indices of Agricultural Products” (APRI) dataset of Eurostat, which we linearly interpolate to obtain monthly observations.³⁰ We do not have data on agricultural industrial production as Euro-

²⁶The results reported in section 3.2 are robust to an alternative identification strategy based on the first principal component of high-frequency changes in 1-, 3-, 6-, and 12-month OIS rates, rather than relying solely on the three-month OIS rate (Jarociński and Karadi, 2020).

²⁷More details on STS data can be found here.

²⁸More precisely, we divide turnover values by the volume of sales (deflated turnover).

²⁹This refers to the statistical classification of products by activity (CPA) code F411X, the main component of sector F41 which covers “new buildings” only.

³⁰The “Price indices of agricultural products (apri pi)” data which can be found here. stat data are only reported at an annual frequency,³¹ so that a monthly interpolation would not be sufficiently accurate.

In STS, data on employment, hours worked, and wages are mainly available at quarterly frequency, and only partly provided at monthly frequency on a voluntary basis by national statistical offices.³² Data on prices, industrial production,

and turnover are available both at monthly and quarterly frequency, depending on the respective country, sector and time period.³³ To maximize the number of observations while preserving the consistency of the dataset, we linearly interpolate quarterly series to monthly and either take the original monthly series or the quarterly interpolation, depending on which of the two has more observations.

For all indices, we use seasonally adjusted series and perform a three-step cleaning procedure.³⁴ First, we exclude all observations of a specific index variable reported with a value of zero. Second, we drop all observations of a series with exactly identical data entries for more than six subsequent months (if the series is monthly) or more than four quarters (if the series is quarterly). Third, we drop entirely data series exhibiting an implausibly high level of volatility or poor data quality, i.e. due to implausibly large discrete jumps at random intervals.

Aggregation from NACE at 88 sectors to 64 sectors

The input-output matrices available from Eurostat are disaggregated at NACE 2 digits, but with few difference compared to the standard NACE 2 digits classification. Indeed, the latter has 88 sectors, while in the input-output tables some sectors are aggregated, resulting in a 64 sectors breakdown. Table A.2 shows the sectors which are aggregated in the input-output tables.

Since STS data are at 88 sectors, we aggregate the variables contained in STS (e.g. PPI, industrial production, etc.) to match the 64 sectors breakdown of the input-output matrices. For sectors B00, F00 and I00 we don't perform any aggregation, since variables are already available also at 1 digit. For the other sectors listed in table A.2 we perform the aggregation calculating a weighted average of the sub

³¹See the "Economic accounts for agriculture (aact eaa)" data which can be found here.

³²See STS regulations here.

³³This implies that for a given country-sector pair we may be able to retrieve only monthly or only quarterly observations, or both. Moreover, monthly and quarterly observations, if both available, are not perfectly consistent at all times in STS.

³⁴We derive seasonally adjusted series in cases for which the seasonal adjustment is not directly carried out by Eurostat by performing a LOESS transformation.

NACE code (88 sectors)	NACE code (64 sectors)
------------------------	------------------------

B05	B00
-----	-----

B06	B00
-----	-----

B07	B00
-----	-----

B08	B00
-----	-----

B09B00
 C10C10-12
 C11C10-12
 C12C10-12
 C13C13-15
 C14C13-15
 C15C13-15
 C31C31 32
 C32C31 32
 F41F00
 F42F00
 F43F00
 I55I00
 I56I00
 J59J59 60
 J60J59 60
 J62J62 63
 J63J62 63
 M69M69 70
 M70M69 70
 M74M74 75
 M75M74 75
 N80N80-82
 N81N80-82
 N82N80-82

Table A.2: Mapping of NACE code 88 to NACE code 64.

sectors, using as weights the “Production value” (variable code V12120) of each sector obtained from the Eurostat Structural Business Statistics (SBS) dataset. SBS describes the detailed structure, economic activity, and performance of businesses over time and covers the countries and sectors of our interest. However, SBS only goes from 2005 to 2020, with some gaps and some countries not reporting in all the years. To have a complete series of weights we perform the following steps. For each country and sector:

1. we linearly interpolate missing values to fill gaps in the time series;
2. we extrapolate missing values filling forward (after the latest observation) and

backward (before the earliest observation), effectively assuming a constant value for the latest available observation;

3. in case a given country-sector is completely missing, we assume its weight is zero;

4. if in a given country all "subsectors" (e.g. C13, C14, C15) that are part of a given aggregation at 64 sector (e.g. C13-C15) are missing, then we assume that they have the same weight (i.e. we make a simple average). In practice, this happens only in the case of Luxembourg for sectors C13, C14, C15.

Finally, for each sector in the FIGARO tables, we compute the share of input sourced outside the production network as the sum of labor, value added and taxes (i.e. rows "W2 D21X31 - Taxes less subsidies on products", "W2 OP NRES - Purchases of non-residents in the domestic territory", "W2 OP RES - Direct purchase abroad by residents", "W2 D1 - Compensation of employees", "W2 D29X39 - Other net taxes on production", "W2 B2A3G - Gross operating surplus") divided by the total input cost of production. Analogously, for each sector, we compute the share of output going to final consumption as the sum of government and household consumption, capital formation and inventories (i.e. columns "P3 S13 - Final consumption expenditure of general government", "P3 S14 - Final consumption expenditure of households", "P3 S15 - Final consumption expenditure of non-profit institutions serving households", "P51G - Gross fixed capital formation", "P5M - Changes in inventories and acquisition less disposals of valuables") divided by total output.

Table A.3: Summary statistics

UnitFreq.Obs.MeanSt. dev.Median

Euro area level

EUR/USD exchange rateM3121.180.161.17

Composite 10y sov. bond yield%M3123.031.603.41

IMF Commodity IndexidxM256134.0838.25126.39

CISSp.n.M3120.180.150.12

3-month OIS%M3051.471.750.78

Monetary policy shockbpsM3120.162.670.00

Country level

Unemployment%M7,1158.794.147.98

HICPidxM6,40995.5115.8197.96

Country-sector level

Producer priceidxM145,98652.8148.0174.04

Industrial productionidxM197,95184.8388.0992.59

TurnoveridxM220,521101.232533.4683.60

EmploymentidxM98,71042.7753.980.00

Total leverageratioA249,1440.590.160.60

Working capitalratioA249,1440.150.150.14

Sources: Altavilla et al. (2025), ECB, Eurostat, IMF, Orbis.

Notes: "idx", "bps" and "p.n." stand for index, basis points and pure number, while M and A denote monthly and annual frequency.

Table A.4: Summary of notation and definitions

Object	Formula	Definition
--------	---------	------------

Upstream effect	From supplier	
-----------------	---------------	--

Transmission from upstream supplier j to downstream customer k , capturing the propagation of cost shocks along the production network.		
---	--	--

Downstream effect	From customer	
-------------------	---------------	--

Transmission from downstream customer k to upstream supplier j , capturing the propagation of demand shocks along the production network.		
---	--	--

$X = [x_{c,d}$

Input-output

$j,k]$ Bilateral flows of intermediate goods and services, matrix

where each element denotes sales from sector j in country c to sector k in country d .

k,d $x_{c,d}$

Total output y_c

$j = P$

$j,k + c_j$ Total output of sector j in country c , defined as the sum of intermediate sales to all countries (d) and sectors (k), as well as final demand.

$x_{c,d}$

$j,k / y_c$

Direct-coefficients

$\Omega =$

Matrix of input coefficients, where each element

j

matrix

measures the share of sector j 's output sold to

sector k .

Leontief inverse $Z = (I - \Omega)^{-1}$ Matrix summarizing the total (direct and indi-

rect) effects of a demand shock propagating from

customers to suppliers via production linkages.

$\tilde{Z} = (I - \Omega^T)^{-1}$ Matrix capturing the total (direct and indirect)

Leontief inverse

(transposed-based)

effects of a cost shock propagating from suppliers

to customers via production linkages.

B Impulse response scaling

We scale the size of a monetary policy tightening shock st in model 8 to imply a peak

increase in the 3m OIS rate – the market-based monetary policy rate proxy from

which monetary policy shocks are identified – by 25 basis points in the first year

after the shock. We then scale the impulse response functions for the macroeconomic

variables of interest to be consistent with such a 25bps peak impact monetary policy

tightening shock. To do so, we proceed in two steps. First, we estimate a euro area

local projection model including broadly the same control variables as in the country-

sector panel model 8, but at the aggregate level, to account for the fact that the

dependent variable y_t is observed at the euro area level only in this setting. We

then derive a scaling parameter $\tau \equiv 0.25$

ψ , with ψ referring to the peak of the impulse

response of the OIS 3m rate to a monetary policy tightening shock within the first

year after the shock, expressed in percentage points. We finally use τ as a scaling

parameter in the impulse response functions of the dependent variable of interest

shown in section 3.2.

The euro area aggregate model is given by:

LX

LX

$$y_{t+h} = \beta_h$$

$$\delta_h K_{t-l} +$$

$$\eta_h X_{t-l} + \epsilon_{t+h}(B.1)$$

$$1 \text{ st } +$$

$$l=1$$

$$l=0$$

"

#

ytst

with $h = 1, 2, \dots, H$, $K_t =$

with vector K_t indeed collecting lags of the dependent variable y_t and of the shock st . Matrix X_t contains the contemporaneous values and lags of the same set of macro-financial control variables as included in model 8, i.e. a GDP-weighted 10y composite euro area sovereign bond yield, the euro-dollar exchange rate, and log-levels of the euro area Composite Indicator of Systemic Stress (CISS), the IMF Commodities Price Index, the euro area harmonized index of consumer prices (HICP) and the euro area unemployment rate. It also includes our main variables of interest, industrial production and producer prices, now measured at the euro area aggregate level.

Figure B.1 shows the impulse response function to a monetary policy tightening shock as obtained from model B.1. Without scaling, the shock refers a one percentage point increase in the shock series from the mean. Given that the mean monetary policy shock in our sample only amounts to 0.2 basis points, the shock impact as 1 in equation B.1 and shown in figure B.1 turns out large.³⁵ Within measured by β_h

the first year, the peak increase in the three-month OIS rate is reached three months after the shock and amounts to 3.2 percentage points. This implies a the scaling parameter $\tau \approx 0.078$.

Figure B.1: Impulse responses to monetary policy tightening shock - aggregate model

3m OIS90 % CI

5

Percentage points

0

-5

0612182430364248Months

Notes: Impulse responses to a monetary policy tightening shock. Estimates for model B.1 estimated on aggregate euro area data with Newey and West (1987) standard errors.

35As discussed in section 2, we identify monetary policy shocks by applying the Jarociński and Karadi (2020) “poor man’s” sign restrictions to the innovations in the 3m OIS rate around policy events as identified by Altavilla et al. (2025).

CVariance decomposition of monetary policy effects

This appendix describes the covariance-based decomposition used to quantify the contribution of different transmission channels to the variance of the monetary policy effect. It also discusses robustness of the obtained variance shares to a sequential Shapley (1953)-style decomposition.

C.1 Covariance-based decomposition

C.1.1 Model-implied components

We start by deriving the monetary policy-related fitted value of the outcome variable obtained from the local projection model

X

$\hat{y}_{MP}$

$t =$

$C_{k,t}, (C.1)$

$k \in K$

where $C_{k,t}$ denotes the contribution of component k associated with the monetary policy shock st . In our application, the set of components is given by

$K = \{\text{Direct, Upstream, Downstream, Average, Non-network}\},$

captured by the coefficients β_1 to β_6 in model 8. Each component aggregates the effects of contemporaneous and lagged interactions between the monetary policy shock and the corresponding transmission channel, as described in the empirical model. The monetary policy-only fitted value $\hat{y}_{MP}$

t therefore captures the total

contribution of the monetary policy shock to the dependent variable, abstracting from other controls and fixed effects.

C.1.2 Variance decomposition

The total variance of the monetary policy-induced component is given by

X

!

$\text{Var}(\hat{y}_{MP}$

$t) = \text{Var}$

$.(C.2)$

$C_{k,t}$

$k \in K$

Using the identity

X

!

X

$\text{Cov}(C_{k,t}, \hat{y}_{MP}$

Var

$C_{k,t}$

$=$

$t), (C.3)$

k

k

we can decompose total variance into the sum of covariances between each component and the overall monetary policy-related fitted value $\hat{y}_{MP}$
 t .

We define the variance share of component k as

$S_k = \text{Cov}(C_{k,t}, \hat{y}_{MP}$

$t) \text{Var}(\hat{y}_{MP}$

$t). (C.4)$

In the empirical implementation, these objects are estimated using sample moments:

TX

$t) = 1T - 1$

$d\text{Var}(\hat{y}_{MP}$

$(\hat{y}_{MP}$

$t - \bar{y}_{MP})^2, (C.5)$

$t = 1$

TX

$t) = 1T - 1$

$(C_{k,t} - \bar{C}_k)(\hat{y}_{MP}$

$$dCov(C_{k,t}, \hat{y}_{MP}$$

$$t - \hat{y}_{MP}). \quad (C.6)$$

$$t=1$$

By construction,

$$X$$

$$S_k = 1. \quad (C.7)$$

$$k \in K$$

The variance share S_k measures the contribution of channel k to fluctuations in the monetary policy effect. A larger value of S_k indicates that variation in component $C_{k,t}$ is strongly aligned with variation in the overall monetary policy-induced response. Conversely, a small value indicates that the component contributes little to overall fluctuations.

Because the components are not orthogonal, the decomposition accounts for their joint comovement with the overall monetary policy-related fitted value $\hat{y}_{MP}$ rather

than isolating purely independent contributions. As a result, the shares reflect both the magnitude of each component and its correlation with other channels.

In particular, a component that is highly volatile but weakly correlated with the overall response will receive a smaller share than a component whose movements closely track the total effect.

Importantly, all components are constructed using the estimated coefficients and all included lags of the monetary policy shock and its interactions. The decomposition therefore captures the full dynamic propagation of monetary policy through the different transmission channels implied by the model.

C.2 Robustness: Sequential variance decomposition

As an alternative approach to the above covariance-based decomposition, we implement a sequential variance decomposition based on the model-implied components broadly following Shapley (1953). Specifically, we again construct the fitted value associated with the monetary policy shock, $\hat{y}_{MP}$

t , as the sum of its constituent com-

ponents as in equation C.1. We then evaluate the incremental contribution of each component to the variance of the fitted values. This approach captures how much each transmission channel contributes to overall fluctuations, conditional on previously included components.

C.2.1 Variance decomposition

To assess the contribution of each component to the variance of $\hat{y}_{MP}$

at t , we construct

a sequence of partial fitted values. Let $\Xi = (k_1, k_2, \dots, k_K)$ denote an ordering of the components. Defining

m_X

$\hat{y}_{MP,(m)}$

at

$C_{k_j,t,m} = 1, \dots, K, (C.8)$

$j=1$

with $\hat{y}_{MP,(0)}$

at $t=0$, the marginal contribution of component k_m is then given by

$\Delta V_{k_m} = \text{Var}(\hat{y}_{MP,(m)} - \hat{y}_{MP,(m-1)})$

$\Delta V_{k_m} = \text{Var}(\hat{y}_{MP,(m)})$

at t . (C.9)

The total variance is defined as

$V = \text{Var}(\hat{y}_{MP,t})$

(C.10)

and the variance share of component k_m is

$S_{k_m} = \Delta V_{k_m}$

V . (C.11)

By construction, the shares again satisfy

$\sum_k S_k = 1$

(C.12)

$k \in K$

As above, the variance share S_k measures how much component k contributes

to overall fluctuations in the monetary policy-related fitted value $\hat{y}_{MP,t}$

when added

to the set of previously included components. Figure C.1 shows that the respective

patterns for variance shares are qualitatively consistent with the ones obtained by

the covariance-based approach reported in figure 6 in the main text.

Figure C.1: Sequential variance decomposition of monetary policy channels

PPI

Industrial production

1.5

1

1

.5

Share

Share

.5

0

-.5

0

-1

0612182430364248Months

0612182430364248Months

Direct FF (β_1) Upstream FF (β_2) Downstream FF (β_3) Average Effect (β_6) Non-network ($\beta_4 + \beta_5$)

Notes: Sequential Shapley (1953)-type variance decomposition of monetary policy-related components in model 8.

C.2.2 Ordering of components

As components may be correlated, the ordering Ξ affects the allocation of shared variation across components. The reported decomposition therefore reflects one economically motivated ordering of transmission channels and can be interpreted as an approximation to a full Shapley (1953) decomposition, which averages over all possible orderings.

In the sequential variance decomposition in figure C.1, we adopt the following ordering:

$$\Xi = (\text{Average, Direct, Upstream, Downstream, Non-network}). \quad (\text{C.13})$$

This ordering reflects an economically motivated hierarchy of transmission channels. We first include the average effect of a monetary policy shock, i.e. barring network-induced nonlinearities, given by β_6 in model 8. We then straightforwardly add the direct effect governed by β_1 and upstream and downstream network effects given by β_2 and β_3 , and finally the non-network related interactions described by β_4 and β_5 . Under this ordering, each component's contribution is interpreted as the incremental increase in the variance of the fitted values when the component is added to the set of previously included components. Figure C.2 shows respective results for the reversed ordering

$$\tilde{\Xi} = (\text{Non-network, Downstream, Upstream, Direct, Average}). \quad (\text{C.14})$$

Results remain broadly robust under complete reversal to those obtained under both the covariance-based and main sequential decomposition as shown in figures 6

and C.1, with the average, downstream financial frictions and non-network effects still explaining the largest shares of the variation in producer prices and industrial production.

Figure C.2: Sequential variance decomposition of monetary policy channels - reverse ordering

PPI

Industrial production

1

1.5

1

.5

.5

Share

Share

0

0

-.5

-1

-.5

0612182430364248Months

0612182430364248Months

Direct FF (β_1) Upstream FF (β_2) Downstream FF (β_3) Average Effect (β_6) Non-network ($\beta_4 + \beta_5$)

Notes: Sequential Shapley (1953)-type variance decomposition of monetary policy-related components in model 8, based on reverse ordering C.14.

D Robustness checks

In the following, we provide robustness checks to the results obtained with empirical model presented in section 3.2. We particularly assess the robustness of our main empirical results in figures 3 to 5 to using an alternative financial frictions measure, namely firms' working capital, to using different means of aggregating firm-level leverage data to the sectoral level, and to running different model specifications.

D.1 Financial frictions measures

In this section, we show that our results remain broadly robust when using an alternative empirical financial frictions measure for ϕ in equations 7 and 8. Figures D.1 to D.3 show the same set of impulse response as depicted in figures 3 to 5 when sector-aggregates of firm level data on the working capital share, defined as working capital expenses/total assets instead of total leverage is used. We assess robustness

to working capital also in light of its importance in the theoretical analysis we carry out in section 4. Overall, results are robust to using the working capital share, with results still indicating the same diverging effects of up- and downstream financial frictions in the transmission of monetary policy over the medium term as displayed in section 3.2. However, using the working capital share results in a slight tilting of results in the short term, with downstream demand effects potentially dominating direct financial constraints effects in the first year after the shock. As shown in figure D.2, the direct financial frictions effect is negative in the first year after the shock, before upstream cost effects become the dominant driver of direct financial constraints effects as in the baseline specification (figure 4). This in turn results in a muting of the inflationary short-term financial constraints effects reported in the baseline results when relying on working capital shares instead of total leverage (figure 3 vs figure D.1).

We also test for differences in aggregating firm-level data on financial frictions measures to the sectoral level. While main results in figures 3 to 5 were derived using sales-based weighted averages of firm level data to generate sectoral measures, figures D.4 to D.6 show that results are robust to changing aggregation such that sectoral levels of total leverage reflect the median firm's leverage holdings.

Finally, we adjust our baseline financial frictions measure in equation 7 to proxy for additional possible distortions requiring a differentiation between cost- and revenue-based input–output matrices as in Baqaee and Farhi (2020). Our main

Figure D.1: Impulse responses to monetary policy tightening shock - working capital

PPI

Industrial production

2

5

1

Percent

Percent

0

0

-1

-5

-2

0612182430364248Months

0612182430364248Months

Average Effect (β_6) Financial Frictions ($\beta_1 + \beta_2 + \beta_3$) 90 % Confidence Interval 90 % Confidence Interval 68 % Confidence Interval 68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

empirical measures in equations 7 follow the theoretical model in assuming perfect competition and no further distortions beyond the financial friction introduced in equation 10. Our empirical and theoretical measures are thus consistent with the approach in Acemoglu et al. (2012), in which, as a result of the absence of other frictions, cost and revenue shares would ultimately coincide, such that both the cost- and revenue based Leontief inverses Z and $\tilde{Z}$ would be based on the same matrix Ω , as stated in table A.4. We thus also run our baseline regression 8 using versions of the financial frictions measures 7 adjusted for additional frictions as in Baqaee and Farhi (2020). However, as we are not able to empirically observe the actual cost-based input-output matrix defined by the elasticity of sector j 's marginal cost with respect to the price of sector k as prescribed by the theoretical setup in Baqaee and Farhi (2020), we approximate the distortion component by using observed input cost shares as reported in input-output tables. Concretely, in comparison to the definitions in table A.4, we now differentiate between a cost-share matrix and a revenue-share matrix, with the two matrices being given by

Figure D.2: Impulse responses to monetary policy tightening shock - direct vs. indirect financial frictions effects - working capital

PPI

Industrial production

2

2

1

1

0

0

Percent

Percent

-1

-1

-2

-2

-3

-3

0612182430364248Months

0612182430364248Months

Direct FF (β_1) Indirect FF ($\beta_2 + \beta_3$) Total FF ($\beta_1 + \beta_2 + \beta_3$) 90 % Confidence Interval 90 % Confidence Interval 68 % Confidence Interval 68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

"

#

xc,d

ijP

$\Omega =$

(D.1)

j,d xd,c

kj + lj

xc,d

"

#

jkyc

$\tilde{\Omega} =$

.(D.2)

j

The two matrices differ in terms of the denominators, where input costs are

k,d xc,d

reported in the denominator of Ω , while total revenue yc

j = P

jk + cj = P

kj + lj + vj + tj is reported in the denominator of matrix $\tilde{\Omega}$. Thus, the

j,d xd,c

revenue-share matrix $\tilde{\Omega}$ accounts also for value added and tax terms vj and tj, which determine our input-output based proxies for additional distortions such as market power and distortive tax policies. Consistently, the adjusted cost- and revenue-based Leontief inverses are now given by:

$$Z = (I - \Omega T)^{-1}(D.3)$$

$$\tilde{Z} = (I - \tilde{\Omega})^{-1}(D.4)$$

Figure D.3: Impulse responses to monetary policy tightening shock - upstream vs. downstream financial frictions effects - working capital

PPI

Industrial production

2

2

1

0

0

Percent

Percent

-1

-2

-2

-4

-3

0 6 12 18 24 30 36 42 48 Months

0 6 12 18 24 30 36 42 48 Months

Upstream FF (β_2) Downstream FF (β_3) Indirect FF ($\beta_2 + \beta_3$) 90 % Confidence Interval 90 % Confidence Interval 68 % Confidence Interval 68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

As shown in figures D.7 to D.9, our main results in figures 3 to 5 are robust to those obtained when estimating model 8 with the adjusted cost- and revenue-based Leontief inverses.

D.2 Model specification

Finally, we also assess the robustness of our results across different specifications of the model, beyond the choice of financial frictions measures and the representation of input-output linkages. In particular, we assess whether estimating the model in level terms instead of long-differences. The level variant of the long-difference model 8 is given by:

Figure D.4: Impulse responses to monetary policy tightening shock - median level of sectoral total leverage

PPI

Industrial production

2

5

1

Percent

Percent

0

0

-1

-5

-2

0612182430364248Months

0612182430364248Months

Average Effect (β_6) Financial Frictions ($\beta_1 + \beta_2 + \beta_3$) 90 % Confidence Interval 90 % Confidence Interval 68 % Confidence Interval 68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

Upstream effect z_t β_h

Downstream effect z_t β_h

$3 \sim \Phi'_{jc,t12-1} \times st\{z\}$ Indirect financial frictions

$2 \Phi'_{jc,t12-1} \times st +$

$y_{jc,t+h} = \beta_h$

$1 \phi_{jc,t12-1} \times st\{z\}$ Direct financial frictions

+

+(D.5)

effect

effect

β_h

$4 a_{jc,t12-1} \times st + \beta_h 5 \sim a_{jc,t12-1} \times st + \beta_h 6 st\{z\}$ Non-network effect

LX

LX

LX

$y_{hHt-l} +$

$\delta_{hKt-l} +$

$$\eta h X_{t-1} + \varphi_j c + \theta m + \kappa t + h + \epsilon_{jc,t+h}$$

+

$$l=0$$
$$I=1$$
$$l=0$$

with $h = 1, 2, \dots, \text{HandL} = 3$

 $y_{jc,t} \phi_{jc,t}^{12} \times st$ 
$$a_{jc,t12} \sim a_{jc,t12} \phi_{jc,t12} \Phi'$$

$$a_{jc,t12} \times st \sim a_{jc,t12} \times st \Phi'$$
$$H_t =$$
$$, K_t =$$
$$j_{C,t12} \times st^{\sim} \Phi'$$
$$j_{c,t12} \sim \Phi'$$
$$jc_{t12} \times st$$

jc,t12

st

Figure D.5: Impulse responses to monetary policy tightening shock - direct vs. indirect financial frictions effects - median level of sectoral total leverage

PPI

Industrial production

2

2

1

1

0

0

Percent

Percent

-1
-1
-2
-2
-3
-3

0612182430364248Months
0612182430364248Months

Direct FF (β_1)Indirect FF ($\beta_2 + \beta_3$)Total FF ($\beta_1 + \beta_2 + \beta_3$)90 % Confidence Interval90 % Confidence Interval68 % Confidence Interval68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel. Compared to the long-difference variant, we include country-sector fixed effects φ_{ic} , which are not present in the differenced version of the model. Jord`a and Taylor (2024) suggest using long-differences to mitigate concerns regarding small sample biases, and comparing results in figures D.10 to D.12 with our main results in figures 3 to 5 confirm that our findings would be broadly robust to such concerns when estimating the model in levels.

Figure D.6: Impulse responses to monetary policy tightening shock - upstream vs. downstream financial frictions effects - median level of sectoral total leverage

PPI

Industrial production

2
2
1
0

Percent
Percent

0
-2
-1
-4
-2

0612182430364248Months

0612182430364248Months

Upstream FF (β_2)Downstream FF (β_3)Indirect FF ($\beta_2 + \beta_3$)90 % Confidence Interval90 % Confidence Interval68 % Confidence Interval68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

Figure D.7: Impulse responses to monetary policy tightening shock - distortion-adjusted Leontief inverses

PPI

Industrial production

2

5

1

Percent

Percent

0

0

-1

-5

-2

0612182430364248Months

0612182430364248Months

Average Effect (β_6)Financial Frictions ($\beta_1 + \beta_2 + \beta_3$)90 % Confidence Interval90 % Confidence Interval68 % Confidence Interval68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

Figure D.8: Impulse responses to monetary policy tightening shock - direct vs. indirect financial frictions effects - distortion-adjusted Leontief inverses

PPI

Industrial production

1

3

2

0

1

Percent

Percent

0

-1

-1

-2

-2

0612182430364248Months

0612182430364248Months

Direct FF (β_1) Indirect FF ($\beta_2 + \beta_3$) Total FF ($\beta_1 + \beta_2 + \beta_3$) 90 % Confidence Interval 90 % Confidence Interval 68 % Confidence Interval 68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

Figure D.9: Impulse responses to monetary policy tightening shock - upstream vs. downstream financial frictions effects - distortion-adjusted Leontief inverses

PPI

Industrial production

2

2

1

1

0

Percent

Percent

0

-1

-1

-2

-3

-2

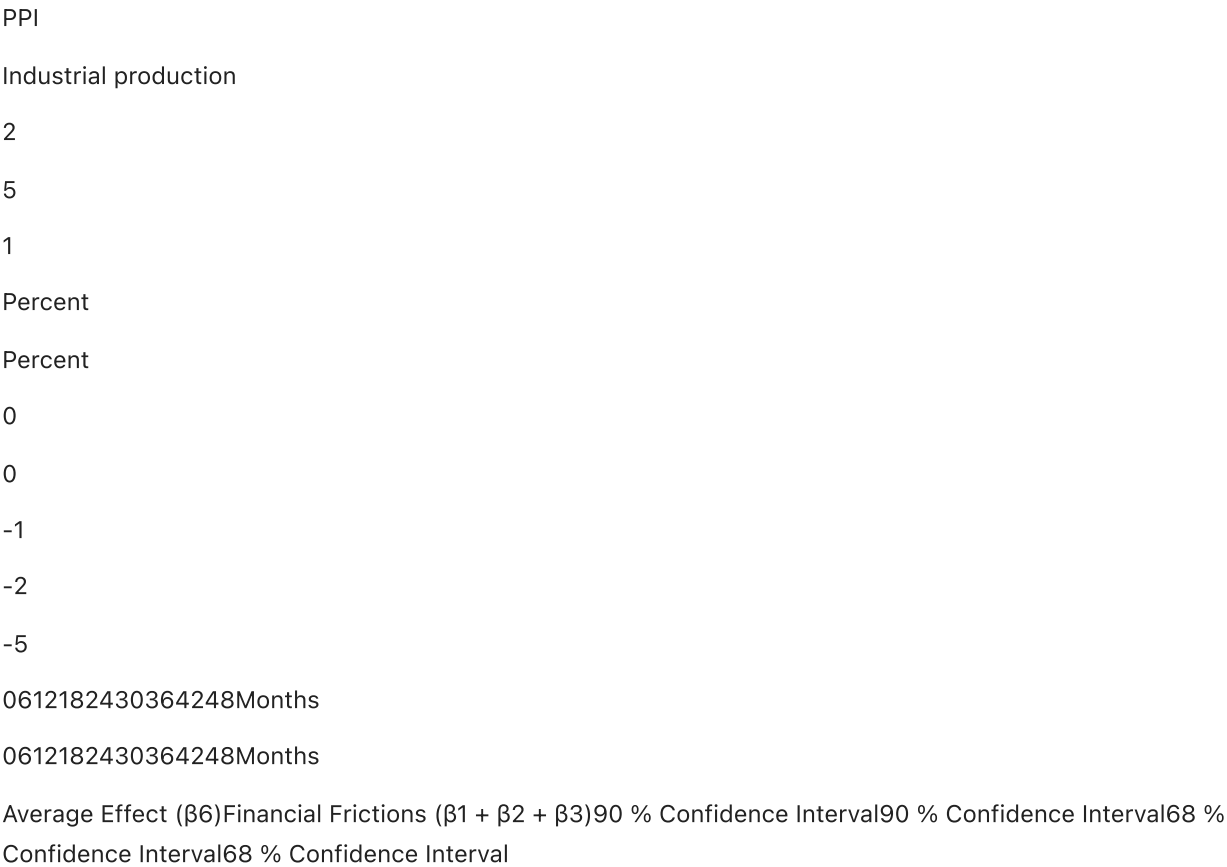
0612182430364248Months

0612182430364248Months

Upstream FF (β_2) Downstream FF (β_3) Indirect FF ($\beta_2 + \beta_3$) 90 % Confidence Interval 90 % Confidence Interval 68 % Confidence Interval 68 % Confidence Interval

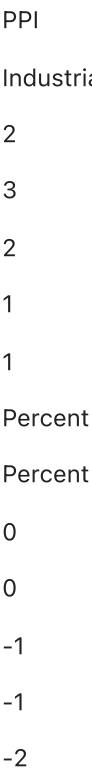
Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

Figure D.10: Impulse responses to monetary policy tightening shock - level specification



Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

Figure D.11: Impulse responses to monetary policy tightening shock - direct vs. indirect financial frictions effects - level specification



-2

0612182430364248Months

0612182430364248Months

Direct FF (β_1) Indirect FF ($\beta_2 + \beta_3$) Total FF ($\beta_1 + \beta_2 + \beta_3$) 90 % Confidence Interval 90 % Confidence Interval 68 % Confidence Interval 68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

Figure D.12: Impulse responses to monetary policy tightening shock - upstream vs. downstream financial frictions effects - level specification

PPI

Industrial production

4

1.5

1

2

.5

Percent

Percent

0

0

-.5

-2

-1

-4

0612182430364248Months

0612182430364248Months

Upstream FF (β_2) Downstream FF (β_3) Indirect FF ($\beta_2 + \beta_3$) 90 % Confidence Interval 90 % Confidence Interval 68 % Confidence Interval 68 % Confidence Interval

Notes: Impulse responses to a monetary policy tightening shock scaled to a peak increase in the 3m OIS rate of 25bp. Estimates for model 8 estimated on the full country-sector euro area panel.

EModel derivations

This appendix provides the complete set of derivations for results stated in section 4.

We first report the firm's optimization problem and the market clearing conditions for completeness, before describing in more detail the closed-form price decomposition 24, the corresponding response to productivity shocks, and the leading-order

proxy mapping of section 4.4. All results are derived at $\sigma = 1$, i.e. with Cobb–Douglas demand, unless otherwise stated.³⁶

We use the notation $\hat{x} \equiv d \log x / di$ for semi-elasticities, and define two friction derivatives recalled throughout the derivations:

$$f_j \equiv \phi_j (1 + i\phi_j) = d \log(1 + i\phi_j) / di$$

$$d_i, g_j \equiv \phi_j (1 + i\phi_j)^2 = f_j (1 + i\phi_j). \quad (\text{E.1})$$

Both f_j and g_j collapse to ϕ_j at $i = 0$. The quantity f_j governs the rate at which revenue shares shrink with i (see section E.4), while g_j appears in intermediate steps, without entering the final price decomposition. Throughout, $\alpha = (\alpha_1, \dots, \alpha_n)^\top$ denotes the vector of labor shares and $D\alpha \equiv \text{diag}(\alpha_1, \dots, \alpha_n)$ denotes the corresponding diagonal matrix.

E.1 Firm optimization and optimality conditions

Cost minimization problem

Sector j takes output prices $\{p_k\}$, the wage w_j , the interest rate i , and its friction parameter ϕ_j as given. It chooses labor l_j and intermediate inputs $\{x_{kj}\}_{k=1}^n$ to

minimize the total cost of producing a given output y_j , subject to the production technology 9.

The financing wedge $1 + i\phi_j$ is a multiplicative markup on all variable inputs, thus not affecting relative input prices. The cost minimization problem is therefore given by:

$$\min_{l_j, x_{kj}} C_j$$

$$\text{s.t. } Y_j = y_j$$

$$\omega_k x_{kj} \geq y_j. \quad (\text{E.2})$$

$$\alpha_j l_j$$

$$\min \{l_j, x_{kj}\} (1 + i\phi_j)$$

$$w_j l_j +$$

$$p_k x_{kj}$$

$$\text{s.t. } \tilde{a}_j l_j$$

$$x_{kj}$$

$$k=1$$

$$k=1$$

³⁶We discuss the revenue equation for general levels of σ in section E.6 below.

As the financing wedge is a common scalar multiplier, the first-order conditions for cost minimization are identical to those of a standard Cobb–Douglas problem without financing frictions. The Lagrangian is given by

$$\omega_k x_k^{\alpha_k} - y_j$$

$$\alpha_j$$

$$X$$

$$Y$$

$$L = (1 + i\phi_j)$$

$$-\mu_j$$

$$w_j l_j +$$

$$p_k x_k$$

$$\tilde{\alpha}_j l$$

$$x$$

$$, (E.3)$$

$$k$$

$$k$$

where μ_j is the Lagrange multiplier on the output constraint. The first-order conditions are:

$$\partial L / \partial l_j$$

$$y_j$$

$$= 0 \Rightarrow (1 + i\phi_j) w_j = \mu_j \alpha_j$$

$$, (E.4)$$

$$l_j$$

$$\partial L / \partial x_k$$

$$y_j x_k$$

$$= 0 \forall k \Rightarrow (1 + i\phi_j) p_k = \mu_j \omega_k$$

$$. (E.5)$$

Dividing equation E.4 by equation E.5 eliminates μ_j :

$$w_j l_j p_k x_k$$

$$= \alpha_j \omega_k$$

$$\forall k. (E.6)$$

This is the standard Cobb–Douglas optimal input ratio showing that the cost shares are exactly equal to the technology exponents, confirming that ω_{kj} is both the exponent in equation 9 and the cost share defined in equation E.6.

Non-financial marginal cost

From the first-order condition E.5, the Lagrange multiplier satisfies

$$\mu_j = (1 + i\phi_j) p_k x_{kj} / (\omega_{kj} y_j) \quad (\text{E.7})$$

for any k . Using the cost shares and summing over all inputs plus labor, the non-financial total cost is

$$X = \sum_j w_j l_j + \sum_k p_k x_{kj} = \tilde{m}c_j y_j, \quad (\text{E.8})$$

where

the non-financial marginal cost is

$$\tilde{m}c_j = \mu_j^{1+\phi_j} = \alpha_j \omega_{kj}^{\alpha_j} \alpha_j \quad (\text{E.9})$$

where

α_j

p

ω

w

$k=1$

$k=1$

Here α_j collects all constant terms from the Shephard-lemma factor demands:

$\omega_{kj} \alpha_j$ arises from the intermediate demand x_{kj}^*

α_j arises from the labor demand l_j^*

the coefficient α

$j = (\alpha_j / w_j) \tilde{m}c_j y_j$, and each ω

$k_j = (\omega_{kj} / p_k) \tilde{m}c_j y_j$.

Optimal pricing under perfect competition

Under perfect competition, firms price at marginal cost. Total marginal cost including the financing wedge is $m c_j = (1 + i\phi_j) \tilde{m}c_j$, so the zero-profit condition gives

n_Y

$$p_j = (1 + i\phi_j) \tilde{m}c_j = 1 + i\phi_j$$

$$\omega_{kj}k. (E.10)$$

$$\alpha_j$$

$$w$$

$$p$$

$$a_j$$

$$k=1$$

This is equation 11 in the main text. Note that the financing wedge $1+i\phi_j$ raises the price proportionally for all levels of output, consistent with perfect competition.

Household optimality

The representative household minimizes expenditure P

$\int p_j c_j$ subject to achieving

utility c from the CES aggregator 13. The first-order conditions give the standard

CES demand system:

$$1-\sigma$$

$$p_j$$

$$p_j c_j = e v_j$$

$$, (E.11)$$

$$p_c$$

$\int p_j c_j$ is total nominal expenditure and $p_c = (P$

$$)^{1/(1-\sigma)}$$
 is

$$\int v_j p^{1-\sigma}$$

where $e = P$

the CES price index. Demand for good j falls with its relative price at elasticity σ ,

and expenditure shares are $p_j c_j / e = v_j (p_j / p_c)^{1-\sigma}$.

E.2 The Cobb–Douglas wage identity

A key identity used throughout the derivations follows directly from the above cost

minimization. From the first-order condition E.4, the labor bill satisfies

$$\mu_j 1 + i\phi_j y_j = \alpha_j \tilde{m}c_j y_j$$

$$w_j l_j = \alpha_j$$

since

$$\tilde{m}c_j = \mu_j 1 + i\phi_j.$$

Since $r_j \equiv p_j y_j = (1 + i\phi_j) \tilde{m}c_j y_j$, we obtain $\tilde{m}c_j y_j = r_j / (1 + i\phi_j)$, and therefore:

$$w_j l_j = \alpha_j r_j 1 + i\phi_j. (E.12)$$

Identity E.12 represents equation 12 in the main text and indicates that the wage bill is a constant fraction α_j of revenue, adjusted downward by the financing wedge. It directly follows from combining the Cobb–Douglas technology fixing cost shares with the working-capital friction driving a wedge between revenue and non-financial cost.

Equation E.12 allows for eliminating the sector wage from the log-price equation.

From labor market clearing $l_j = \bar{l}_j$, we get

$$w_j = \alpha_j r_j (1 + i\phi_j)^{-1} \bar{l}_j$$

, (E.13)

such that $\log w_j = \log \alpha_j + \log r_j - \log(1 + i\phi_j) - \log \bar{l}_j$. This substitution converts the log-price equation from a system in (p_j, w_j) to a system in (p_j, r_j) , where revenues are determined directly by the demand side.

E.3 The log-price equation

In the following, we describe the derivation steps for obtaining equation 20 in the main text.

Step 1: Taking logarithms of the pricing condition

Taking logarithms of equation E.10

n_Y

$$p_j = 1 + i\phi_j$$

$$\omega_k \prod_{k=1}^K p_k. (E.14)$$

$$\alpha_j$$

$$p$$

$$w$$

$$\alpha_j$$

$$k=1$$

yields

n_X

$$\log p_j = \log(1 + i\phi_j) - \log \alpha_j + \alpha_j \log w_j +$$

$$\omega_k \log p_k. (E.15)$$

$$k=1$$

Step 2: Eliminating the sectoral wage

Substituting $\log w_j = \log \alpha_j + \log r_j - \log(1 + i\phi_j) - \log \bar{l}_j$ into equation E.15 yields

$$\log p_j = \log(1 + i\phi_j) - \log \alpha_j$$

n_X

$$\log \alpha_j + \log r_j - \log(1 + i\phi_j) - \log \bar{l}_j$$

$+ \alpha_j$

+

$\omega_{kj} \log p_k$. (E.16)

$k=1$

Collecting the $\log(1 + i\phi_j)$ terms yields

n_X

$\log(1+i\phi_j)+\alpha_j \log r_j+(-\log \alpha_j + \alpha_j \log \alpha_j -\alpha_j \log \tau_j)=c_j$, constant

$\log p_j=(1-\alpha_j)$ net markup weight

+

$\omega_{kj} \log p_k$.

$k=1$

(E.17)

Economically, the weight $1 - \alpha_j$ indicates that an increase in i raises the cost of all variable inputs through the wedge $1 + i\phi_j$. However, the wage identity 12 shows that the wage already adjusts proportionally as revenues and hence the wage bill are determined on the demand side, such that the net cost-push on prices only stems from the non-labor share of inputs, determined by the weight $1 - \alpha_j$.

Step 3: Stacking into a vector system

We define the following vectors: $p = (\log p_j)$, $v = (\log(1 + i\phi_j))$, $r = (\log r_j)$, and $c = (c_j)$. Equation E.17 stacked across all j gives

$p = (I - D\alpha)v + D\alpha r + c + \Omega^T p$. (E.18)

$k \omega_{kj} \log p_k$ becomes $[\Omega^T p]_j$ as ω_{kj} has row = seller

The input-output term P

k and column = customer j , such that Ω^T has row = customer j and $[\Omega^T p]_j = P$

$k \omega_{kj} \log p_k$.

Step 4: Solving for the price vector

Rearranging equation E.18 and premultiplying by $H = (I - \Omega^T)^{-1}$ yields

$(I - D\alpha)v + D\alpha r + c$

$p = H$

(E.19)

which for a single sector k implies

nX

$$(1 - \alpha_j) \log(1 + i\phi_j) + \alpha_j \log r_j + c_j$$

$$\log p_k =$$

$$h_{kj}$$

$$. (E.20)$$

$$j=1$$

This is equation 20 in the main text. The element h_{kj} of the cost Leontief inverse weighs sector j 's contribution to sector k 's price, aggregating all supply-chain paths of arbitrary length connecting j to k upstream.

E.4 The revenue-share matrix and its derivative

Cost and revenue shares

The cost-share matrix Ω has element $\omega_{kj} = p_k x_{kj} / (\sum_j m_{cj} y_j)$, the share of sector j 's non-financial production costs paid to upstream supplier k . The Cobb–Douglas technology and cost minimization imply that this equals the exponent ω_{kj} in equation 9.

The revenue-share matrix $\tilde{\Omega}$ has element $\tilde{\omega}_{kj} = p_k x_{kj} / (p_j y_j)$, the share of sector j 's nominal output sold to sector k as intermediate inputs. Since $p_j y_j = (1 + i\phi_j) \sum_j m_{cj} y_j$, we obtain

$$\tilde{\omega}_{kj} = \omega_{kj} (1 + i\phi_j). (E.21)$$

The financing wedge compresses the revenue share below the cost share whenever $i > 0$. The column sum of $\tilde{\Omega}$ is P

$\sum_k \tilde{\omega}_{kj} = (1 - \alpha_j) / (1 + i\phi_j) < 1 - \alpha_j$, so $\tilde{\Omega}$ is strictly more compressed than Ω .

Derivative of the revenue-share matrix

Differentiating equation E.21 with respect to i yields

$$d\tilde{\omega}_{kj}$$

$$d\tilde{\omega}_{kj} = -\omega_{kj} \phi_j / (1 + i\phi_j)^2 = -\tilde{\omega}_{kj} \times \phi_j / (1 + i\phi_j) = -\tilde{\omega}_{kj} f_j, (E.22)$$

where $f_j = \phi_j / (1 + i\phi_j)$ is the friction derivative defined in equation E.1. The second equality uses $\omega_{kj} / (1 + i\phi_j)^2 = \tilde{\omega}_{kj} \times \phi_j / (1 + i\phi_j)$. In matrix form:

$$d\tilde{\Omega}$$

$$d\tilde{\Omega} = -\tilde{\Omega} \text{diag}(f). (E.23)$$

Importantly, this derivative involves f_j , not $g_j = \phi_j / (1 + i\phi_j)^2$, such that the ratio $-d\tilde{\omega}_{kj} / d\tilde{\omega}_{kj} = \tilde{\omega}_{kj} f_j$, and not $\tilde{\omega}_{kj} g_j \times (1 + i\phi_j)$. The factor $(1 + i\phi_j)$ already

appears in the denominator of $\tilde{\omega}_{kj}$, leaving f_j as the correct multiplier.

E.5 Leontief inverses and derivatives

Definitions

The cost-based Leontief inverse is defined as

∞X

$$H \equiv (I - \Omega^\top)^{-1} =$$

$$(\Omega^\top)^s = I + \Omega^\top + (\Omega^\top)^2 + \dots \quad (\text{E.24})$$

$s=0$

The element h_{kj} counts the total intermediate cost importance of sector j in the production of sector k : At order s , the term $(\Omega^\top)^s$

kj accounts for paths of exactly s

input-output links from k to j upstream. Since Ω has row = supplier and column = customer, $\Omega^\top$ has row = customer and the matrix product chains cost flows from customers back to their suppliers.

The revenue-based Leontief inverse is

$$2 + \dots \quad (\text{E.25})$$

$$Z \equiv (I - \tilde{\Omega})^{-1} = I + \tilde{\Omega} + \tilde{\Omega}^2 + \dots$$

The element z_{kj} captures the propagation of demand shocks from customer j forward to upstream supplier k through revenue-share linkages. Here $\tilde{\Omega}$ is used without transpose because demand flows from customer upstream to suppliers, and in $\tilde{\Omega}$ the column j already describes customer j 's expenditure shares on its suppliers.

At $i = 0$, $\tilde{\Omega} = \Omega$ and the two inverses satisfy $Z^0 = H^\top$

0 up to the row/column

convention, while they diverge for $i > 0$. Moreover, the Cobb–Douglas accounting

identity $a_j + P$

$k \omega_{kj} = 1$ implies

$$\alpha = (I - \Omega^\top)^{-1} \mathbf{1}, H\alpha = \mathbf{1}. \quad (\text{E.26})$$

Derivative of Z with respect to i

We apply the matrix inverse derivative identity, stating that if $A(i)$ is invertible and differentiable in i , then $d(A^{-1})/di = -A^{-1}(dA/di)A^{-1}$, with $A = I - \tilde{\Omega}$ such that

$$di = -Z d(I - \tilde{\Omega})$$

$$di Z = Z d \tilde{\Omega}$$

dZ

$$di Z. \quad (\text{E.27})$$

Substituting equation E.23 yields

dZ

$$d_i = -Z^{-1} \Omega \text{diag}(f) Z. (E.28)$$

Derivative of H with respect to i

The cost-share matrix Ω is a technology parameter and does not depend on i. Thus, matrix H also does not depend on i directly. However, for completeness and given that it enters through the revenue side with $\sigma = 1$, the formula analogous to equation E.28 would be

dH

$$d_i = 0, (E.29)$$

since Ω is fixed. This depicts an important asymmetry between matrices H and

Z: The cost Leontief inverse is invariant to i, whereas the revenue Leontief inverse changes with i as $\tilde{\Omega}$ depends on i through the financing wedge.

E.6 Nominal revenues and the revenue equation

Deriving the revenue equation

Multiplying the goods market clearing condition 14 by p_j gives $p_j y_j = p_j c_j + \sum_k p_j x_{jk}$. Sector j's revenue satisfies

$\sum_k$

$$r_j = p_j c_j +$$

$$\sum_k \omega_{jk} r_k, (E.30)$$

$\sum_{k=1}$

where $\omega_{jk} r_k = p_j x_{jk}$ determines sector j's revenue from selling to customer k

and $p_j c_j$ is household expenditure on good j. Stacking across all j in vector form

implies $r = p \cdot ch + \tilde{\Omega} r$, where $p \cdot ch$ denotes household expenditure by sector. We further obtain

$$r = (I - \tilde{\Omega})^{-1} (p \cdot ch) = Z (p \cdot ch). (E.31)$$

Substituting the household demand system E.11 with $p_c = 1$ gives

$$v \cdot p^{1-\sigma}$$

$$r = Z e$$

$$. (E.32)$$

Simplification at $\sigma = 1$

With $\sigma = 1$, $p^{1-\sigma}$

$$j = p_0$$

$j = 1$ for all j, such that

$$r = e Z v = e p, p \equiv Z v. (E.33)$$

Consequently, prices cancel from the revenue equation entirely, yielding the key

property for making the Cobb–Douglas case with $\sigma = 1$ analytically tractable:

Revenues are proportional to ρ_j , which depends on the interest rate i only through Z and not through prices directly, while prices depend on revenues through equation E.20. Consequently, the system becomes block-triangular, and we solve for ρ and hence revenues before substituting into the price equation.

The case of $\sigma \neq 1$

For $\sigma \neq 1$, differentiating equation E.32 gives

$$\zeta_{jk}^* \hat{p}_k - \beta(\sigma)$$

$\times$

$$\hat{r}_j = \hat{e} + (1 - \sigma)$$

$\hat{p}_j$, (E.34)

k

where ζ_{jk} are revenue-weighted price elasticities and $\beta(\sigma)$

captures a modified

customer-friction exposure. The term $(1 - \sigma) P$

$k \zeta_{jk}^* \hat{p}_k$ reintroduces $\hat{p}$ on the right-

hand side of the revenue equation, creating a genuine simultaneous system without

a closed-form solution for general values of σ . Alternatively, the CES equilibrium

must be solved numerically, as carried out in the simulations shown in appendix F.

E.7 Revenue response to interest rate changes with $\sigma = 1$

Differentiating ρ

Taking logs of equation E.33 gives $\log r_j = \log e + \log \rho_j$, and differentiating with respect to i yields

$$d\rho_j$$

$$\hat{r}_j = \hat{e} + 1$$

$d\hat{p}_j$. (E.35)

ρ_j

We compute $d\rho/di = d(Zv)/di$ using equation E.28 to obtain

$$d\rho$$

$$di = dZ$$

$$d\log v = -Z^{-1} \Omega \text{diag}(f) Z v = -Z^{-1} \Omega \text{diag}(f) \rho. \quad (\text{E.36})$$

Customer-friction exposure

We define the customer-friction exposure of sector j as

$$Z^{-1} \Omega \text{diag}(f) \rho$$

$$\beta_j \equiv -1$$

$$d\rho_j$$

$$j\rho_j$$

$$\geq 0. (E.37)$$

$$d_i =$$

$$\rho_j$$

Writing out the numerator of the last term in equation E.37 explicitly gives

$$[Z^{-1} \Omega \text{diag}(f) \rho]_j =$$

$$\sum_l$$

$$\sum_m$$

$$z_{jl}$$

$$\tilde{\omega}_{lm} f_m p_m. (E.38)$$

$$m$$

$$l$$

The inner sum $\sum_m$

$\tilde{\omega}_{lm} f_m p_m$ measures how quickly sector l 's revenue share is

contracting due to its customers' financing costs: The term $f_m p_m$ defines the rate

at which p_m falls, weighted by how much of sector l 's revenues come from customer

m . The outer Leontief inverse Z propagates this contraction upstream to sector j

through all demand chains of arbitrary length. Dividing by ρ_j , β_j is the proportional

rate at which sector j 's revenue contracts when i rises by one unit. It equals zero if

all of j 's customers (and their customers, etc.) have $\phi_m = 0$.

Substituting back into equation E.35 gives

$$\hat{r}_j = \hat{e} - \beta_j. (E.39)$$

Each sector's revenue semi-elasticity has two components: A common income

effect $\hat{e}$ shared by all sectors, and a sector-specific demand contraction $-\beta_j$ driven

by downstream financial frictions.

E.8 The income semi-elasticity $\hat{e}$

Differentiating the price system

Differentiating equation E.20 with respect to i , and noting that c_j is invariant to i :

$$n \sum_j$$

$$(1 - \alpha_j) f_j + \alpha_j \hat{r}_j$$

$$\hat{p}_k =$$

$$h_{kj}$$

$$\text{.(E.40)}$$

$$j=1$$

Substituting $\hat{r}_j = \hat{e} - \beta_j$ from equation E.39 gives

$$\hat{p} = H$$

$$(I - D\alpha)f - D\alpha\beta$$

$$+ \hat{e} \mathbf{1}, \text{(E.41)}$$

The numéraire condition

Since $p_c = 1$ at all times, differentiating the CES price index with $\sigma = 1$ gives

$$\log p_c = v_T p = \text{const}, \text{ hence } v_T \hat{p} = 0.$$

Solving for $\hat{e}$

Premultiplying equation E.41 by v_T and imposing $v_T \hat{p} = 0$ gives

$$0 = v_T H$$

$$+ \hat{e} v_T \mathbf{1}. \text{(E.42)}$$

$$(I - D\alpha)f - D\alpha\beta$$

Using $v_T \mathbf{1} = 1$, solving for $\hat{e}$ yields equation 23 of the main text:

$$\hat{e} = v_T H$$

$$D\alpha\beta - (I - D\alpha)f$$

$$\text{.(E.43)}$$

Defining $\lambda \equiv H^T v$ as the vector of Domar weights, which measures each sector's total importance in final demand through all supply chains, implies

$$X$$

$$X$$

$$\alpha_j \beta_j - (1 - \alpha_j) f_j$$

$$\hat{e} =$$

$$\lambda_j$$

$$,$$

$$\lambda_j \alpha_j = 1. \quad (\text{E.44})$$

j

j

This expression is the Domar-weighted net effect of monetary tightening on the aggregate price level. The term $(1 - \alpha_j) f_j$ is the positive cost-push component in sector j 's price equation, while $\alpha_j \beta_j$ captures the offsetting demand-contraction component from constrained customers' lower revenues and hence prices. Since the numéraire pins $p_c = 1$, $\hat{e}$ must adjust to balance these two forces in the aggregate. If cost pressures dominate, $\hat{e} < 0$ and aggregate real income falls, dampening demand and stabilizing prices. Crucially, $\hat{e}$ depends on the economy-wide distribution of frictions $\{\phi_j, \beta_j, \alpha_j\}$ through the Domar weights and carries no sector-specific information about sector k 's own suppliers or customers.

E.9 The price-interest rate elasticity decomposition

In the following, we provide the derivations leading to the decomposition 24 in the main text.

For a single sector k , we get from equation E.41

$$n_X$$

$$n_X$$

$$h_{kj}(1 - \alpha_j) f_j -$$

$$\hat{p}_k =$$

$$h_{kj} \alpha_j \beta_j + \hat{e}. \quad (\text{E.45})$$

$$j=1$$

$$j=1$$

We apply the identity $H = I + (H - I)$ to each of the two sums. For any vector

x , this yields

$$X$$

$$X$$

$$(h_{kj} - 1_{j=k}) x_j, \quad (\text{E.46})$$

$$h_{kj} x_j = x_k +$$

j

j

where the first term isolates the own-sector ($j = k$) contribution and the second collects all network ($j \neq k$) effects through paths of length ≥ 1 . This is valid because $s = 1(\Omega^\top)s \geq 0$ element-wise, with the diagonal entries $h_{kk} - 1 \geq 0$.

$$H - I = P_\infty$$

Applying to the cost sum implies

X

X

$$h_{kj}(1 - \alpha_j)f_j = (1 - \alpha_k)f_k +$$

$$(h_{kj} - 1)j=k(1 - \alpha_j)f_j, (E.47)$$

j

j

and applying to the demand sum gives

X

X

$$(h_{kj} - 1)j=k\alpha_j\beta_j. (E.48)$$

$$h_{kj}\alpha_j\beta_j = \alpha_k\beta_k +$$

j

j

Substituting equations E.47 and E.48 into equation E.45, and writing $f_j =$

$\phi_j/(1 + i\phi_j)$ explicitly yields

$(h_{kj} - 1)j=k(1 - \alpha_j)\phi_j1 + i\phi_j|\{z\}$ (ii) Supplier cost

$d_i = (1 - \alpha_k)\phi_k1 + i\phi_k|\{z\}$ (i) Direct cost

$d \log p_k$

X

+

j

(E.49)

X

$-\alpha_k \beta_k |\{z\}$ (iii) Own customers

-

$(h_{kj} - 1)j=k\alpha_j \beta_j$

$+ \hat{e}|\{z\}$ (v) Income

j

$|\{z\}$ (iv) Suppliers'

customers

with β_j and $\hat{e}$ given by equations E.37 and E.43. This is equation 24 in the main

text.

E.10 Price response to productivity shocks

The decomposition above characterizes the reaction of sectoral prices to an interest-

rate shock. It is useful to contrast this with the reaction to a sectoral productivity shock. Let $\text{alog} \equiv (\log a_1, \dots, \log a_n)^\top$ denote the vector of log productivities entering the non-financial marginal cost in equation E.9. Keeping the dependence on productivity explicit, the log-price system can be written as

$$(I - D\alpha)v + D\alpha r - \text{alog} + c_0$$

h

i

$$p = H$$

,(E.50)

where c_0 collects terms that are independent of prices, revenues, interest rates, and productivity. Consider a small productivity change $d\text{alog}$, holding i , ϕ , Ω , and final expenditure weights fixed. Differentiating equation E.50 gives

$$dp = -H d\text{alog} + HD\alpha dr. \quad (\text{E.51})$$

At $\sigma = 1$, the revenue equation is $r_j = ep_j$, with $p = Zv$. Since Z depends on the revenue-share matrix and the financing wedges, but not directly on productivity, p is unchanged by $d\text{alog}$ when i and Ω are held fixed. Therefore

$$dr = d \log e. \quad (\text{E.52})$$

Using $H\alpha = 1$, equation E.51 becomes

$$dp = -H d\text{alog} + d \log e. \quad (\text{E.53})$$

The numéraire condition $p_c = 1$ implies, at $\sigma = 1$, $v^\top dp = 0$. Applying this condition to equation E.53 yields

$$d \log e = v^\top H d\text{alog} = \lambda^\top d\text{alog}, \lambda \equiv H^\top v. \quad (\text{E.54})$$

Thus the full equilibrium price response to an arbitrary vector of productivity shocks is

$$dp = -H d\text{alog} + 1 \lambda^\top d\text{alog}. \quad (\text{E.55})$$

For a productivity shock only in sector k , this gives

$$\partial \log p_j \partial \log a_k$$

$$= -h_{jk} + \lambda_k. \quad (\text{E.56})$$

The first term, $-h_{jk}$, is the cost-network effect: higher productivity in sector k

lowers the unit cost of every sector j that uses k , directly or indirectly, as an input.

The strength of this pass-through is governed by the same cost Leontief inverse that appears in the upstream cost channel of the interest-rate decomposition. The second term, λ_k , is common across all destination sectors j for a given shocked sector k . It is not a technological pass-through term; it is the adjustment in nominal expenditure required to keep the consumption price index fixed at the numéraire $p_c = 1$. Without

imposing this numeraire adjustment, or equivalently holding nominal expenditure fixed, the partial cost effect would simply be

$$\frac{\partial \log p_j}{\partial \log a_k} = -h_{jk}. \quad (\text{E.57})$$

$$d \log e = 0$$

This highlights the difference between productivity and interest-rate shocks. An interest-rate shock changes sectoral prices through the financing wedge f_j , the induced change in revenue shares and downstream customer demand summarized by β_j , and the common income term $\hat{e}$. A productivity shock, by contrast, does not directly affect the financing wedge or the revenue-share matrix at $\sigma = 1$. Its relative-price effect is therefore a pure cost-network propagation, up to the common numeraire adjustment. A sector with a high Domar weight λ_k has a large aggregate effect on the consumption price index; after imposing $p_c = 1$, sectors with weak input dependence on k may therefore experience a positive nominal price response through the common λ_k term even though the underlying cost effect $-h_{jk}$ is non-positive.

E.11 Proof of first-order proxy mappings

In the following, we provide proofs for the mappings presented in section 4.4. We derive each approximation step by step as $i \rightarrow 0$.

Step 1: Cost channels at $i = 0$

At $i = 0$, the friction derivative collapses to $f_{ji}|_{i=0} = \phi_j / (1 + 0) = \phi_j$. Substituting directly into terms (i) and (ii) of equation E.49 yields

$$(i) : (1 - \alpha_k) f_k \rightarrow (1 - \alpha_k) \phi_k, \quad (\text{E.58})$$

$\times$

$\times$

$$(h_{kj} - 1_{j=k})(1 - \alpha_j) f_j \rightarrow$$

$$(h_{kj} - 1_{j=k})(1 - \alpha_j) \phi_j \quad (\text{E.59})$$

(ii) :

j

j

without need for further approximation for the cost channels.

Step 2: Customer channels at $i = 0$

At $i = 0$, $\tilde{\Omega} = \Omega$, such that the revenue-based Leontief inverse reduces to $Z|_{i=0} =$

$Z_0 = (I - \Omega)^{-1}$ and $f_j = \phi_j$. Substituting this into equation E.37 yields

$$i=0 = [Z_0 \Omega \text{diag}(\phi) \rho_0] j$$

β_j

$$\rho_0 = Z_0 v. (E.60)$$

$$\rho_{0,j}$$

We use the Leontief inverse identity $Z_0 \Omega = Z_0 - I$, which follows from $(I - \Omega) Z_0 = I$ rearranged as $Z_0 \Omega = Z_0 (I - (I - \Omega)) = Z_0 - I$ and substitute such that

X

$$[Z_0 \Omega \text{diag}(\phi) \rho_0]_j = [(Z_0 - I) \text{diag}(\phi) \rho_0]_j =$$

$$(z_{0,jm} - 1_{j=m}) \phi_m \rho_{0,m}. (E.61)$$

m

To obtain the proxy $[(Z_0 - I) \phi]_j$, we factor out $\rho_{0,m}$ by using that Domar weights

$\rho_{0,m}$ are approximately equal across sectors at $i = 0$, i.e. $\rho_{0,m} \approx \bar{\rho}_0$ for all m, where

$$\bar{\rho}_0 = v^T Z^T$$

0 1/n:

$$\beta_j |_{i=0} \approx \bar{\rho}_0 \rho_{0,j}$$

$$[(Z_0 - I) \phi]_j \approx [(Z_0 - I) \phi]_j = \tilde{\Phi}_j, (E.62)$$

where the second approximation absorbs $\bar{\rho}_0 / \rho_{0,j}$ into the estimated coefficient as

at leading order, this ratio is approximately constant across sectors and therefore

captured by the regression constant or absorbed by the regression coefficient β_h

3 .

Substituting $\beta_j \approx \tilde{\Phi}_j$ into the combined customer channels gives

$$-[H D \alpha \beta]_k \approx -[H D \alpha (Z_0 - I) \phi]_k. (E.63)$$

Step 3: Income term

The income term is $\hat{e} \times s_t$ because $H \alpha = 1$. The factor $\hat{e}$ depends on economy-wide

averages $\{\phi_j, \beta_j, \alpha_j\}$ through Domar weights (see equation E.44) and is therefore

a scalar common to all sectors in a given period. In the empirical specification,

country-time fixed effects θ_{ct} absorb any term of the form $f(t) \times s_t$ that does not

vary across sectors. Hence the income channel is fully absorbed in the Cobb–Douglas

benchmark.

Step 4: From exact channels to observable proxies

The exact upstream term (ii) is P

$j(h_{kj} - 1_{j=k})(1 - \alpha_j)\phi_j$. The empirical upstream

proxy is $\Phi_k = [(H - I)\phi]_k = P$

$j(h_{kj} - 1_{j=k})\phi_j$. The difference is the weight $(1 - \alpha_j)$.

The proxy does not account for this weight, thereby introducing an approximation

error proportional to the cross-sectional variation in α_j . This approximation is valid

when α_j has limited cross-sectional variation, in which case $1 - \alpha_j$ is approximately

constant across j and absorbed into the estimated coefficient β_h

2 . The approach of

including $\alpha_k x_{st}$ as a control in the empirical specification partially corrects for this, as mentioned in the main text.

For the downstream proxy, the exact term is $-[H\alpha(Z-I)\phi]_k$ at $i = 0$. The empirical proxy $\tilde{\Phi}_k = [(Z-I)\phi]_k$ uses the full revenue-based Leontief inverse $(Z-I)$ rather than just one-step linkages. This choice is justified as the customer channel propagates through all demand chains of arbitrary length via matrices Z and H , and not only stems from direct customers relations alone. The α_k weight and the outer matrix H application are absorbed into the coefficient β_h

3 under the approximation

that α_j is approximately constant.

Sign predictions

The sign predictions follow directly from non-negativity: All entries of matrices $H-I$ and $Z-I$ are non-negative, as Leontief inverses have only non-negative entries and diagonal entries exceeding one, and all $\phi_j \geq 0$ and $\alpha_j \in [0, 1]$. Therefore:

$$(1 - \alpha_k)\phi_k \geq 0 \Rightarrow \beta_h$$

1 > 0 (direct),

$$\Phi_k = [(H-I)\phi]_k \geq 0 \Rightarrow \beta_h$$

2 > 0 (upstream),

$$-[H\alpha \tilde{\Phi}]_k \leq 0 \Rightarrow \beta_h$$

3 < 0 (downstream).

E.12 Regularity Conditions

The derivations above require the following conditions to hold:

1. Cobb–Douglas technology. Constant returns-to-scale Cobb–Douglas production with fixed cost-share parameters $\{\alpha_j, \omega_{kj}\}$ satisfying $\alpha_j + \sum_k \omega_{kj} = 1$

for all j . This ensures the wage identity E.12 holds exactly and that cost shares equal technology exponents.

2. Multiplicative working-capital friction. The friction $(1 + i\phi_j)$ is multiplicative and applies to all variable inputs, with $\phi_j \geq 0$ and $i \geq 0$. This ensures the financing wedge enters the pricing equation in a log-linear way as required for the decomposition.

3. Invertibility of Leontief matrices. Both $(I - \Omega_T)$ and $(I - \tilde{\Omega})$ must be invertible. A sufficient condition is that the spectral radii of Ω_T and $\tilde{\Omega}$ are

both strictly less than one, which is the standard input-output productivity condition ensuring that finite amounts of primary inputs produce finite outputs. Since $\tilde{\Omega} = \Omega \text{diag}((1 + i\phi_j)^{-1}) \leq \Omega$ element-wise, if Ω satisfies the condition, then so does $\tilde{\Omega}$ for all $i \geq 0$.

4. Interior equilibrium. All equilibrium prices, wages, and revenues are strictly positive. This ensures that log-linearization around equilibrium is valid and that the Leontief expansion converges.

5. First-order approximation accuracy. The limits $f_j \rightarrow \phi_j$ and $\tilde{\Omega} \rightarrow \Omega$ as $i \rightarrow 0$ generate approximation errors of order $O(i)$. Additional proxy errors arise from replacing weighted network objects such as $[H(I - D\alpha)\phi]^k$ and $[HD\alpha(Z - I)\phi]^k$ with the observable proxies ϕ^k , Φ^k , and $\tilde{\Phi}^k$. These errors are small when labor shares and Domar weights do not vary too strongly across sectors, as assessed in the simulation evidence in appendix F.

F Simulation-based validation of empirical proxies

This appendix reports the full set of results from the simulation exercise sketched in section 4.5. We simulate $R = 500$ variants of the heterogeneous CES economies prescribed by the model in section 4, with $n = 70$ sectors each. We solve the model equilibrium, and subsequently estimate two regressions, one reflecting the full decomposition in equation 24, and a second one based on the proxy regression 28 mapping the theoretical proxies to the empirical regression model 8.

F.1 Simulation setup

The simulation draws $R = 500$ random economies, each with $n = 70$ sectors. For each economy, we proceed along the following steps.

1. Parameterization

The labor share α_j is drawn uniformly from $[0.3, 0.9]$ for each sector j . The input-output cost-share matrix ω_{kj} (share of seller k in buyer j 's costs) is random and row-stochastic: columns sum to $(1 - \alpha_j)$ so that labor plus intermediate inputs equal unity. Friction parameters ϕ_j are drawn uniformly from $[0, 1]$ to represent the range of sectoral leverage observed in data. Exogenous final demand v is drawn from a uniform distribution on the simplex (after normalization).

2. Equilibrium solution: CES at baseline policy rate

For a baseline policy rate $i_0 = 5\%$, we solve the nonlinear CES equilibrium system via `fsolve` (MATLAB) using a numerical tolerance of 10^{-12} on both the function residual and the change in iterates. The initial guess for `fsolve` comes from the

Cobb-Douglas closed-form solution at i_0 , which is highly accurate and reduces the iteration count substantially. We adopt the CPI numeraire $v \top \log p = 0$ to ensure comparability across all elasticities σ .

3. Price derivative via finite difference

To compute the price response to a small interest rate increase $di = 10^{-6}$, we solve the equilibrium again at $i_1 = i_0 + di$ and compute

$$di \approx \log p_k(i_1) - \log p_k(i_0)$$

$$d \log p_k$$

$$di. (F.1)$$

This finite-difference derivative is used as the dependent variable in Regression

1.

4. Decomposition channels

Using the equilibrium at i_0 , we compute the decomposition channels (direct, supplier, own-customers, suppliers' customers, income) as prescribed by equation 24. These channels are evaluated at the Cobb-Douglas closed-form solution (Step 1), which provides the Leontief inverses H and Z consistent with the CPI numeraire.

5. Shock distribution and multiple observations per economy

To generate cross-sectional variation comparable to the time-series variation of monetary shocks in the empirical panel, we draw $n_{\text{shocks}} = 10$ monetary policy shocks per economy from a normal distribution $si \sim N(0.0025, 0.00152)$, representing mean 25bp and standard deviation 15bp. Shocks are floored at 1bp to avoid negligible perturbations. Each shock is added to the baseline rate: $is = i_0 + si$. We solve the equilibrium at is , compute the price change $\Delta \log p_k = \log p_k(is) - \log p_k(i_0)$, and derive the regressors $(si\phi_k, si[H\phi]_k, si[(Z - I)\phi]_k)$ mirroring the empirical design. This yields approximately $500 \times 10 = 5000$ observations for Regression 2. Observations with negligible price responses ($|\Delta \log p| > 1\%$) or non-finite values are dropped as numerical failures.

Summary

The procedure yields two regression datasets: Regression 1 has ≈ 500 observations, one per economy, the finite-difference derivative and five channels. Regression 2 has ≈ 5000 observations (ten shocks per surviving economy, the price change and three proxy regressors).

F.2 Regression 1: Channel decomposition at $\sigma = 1$

We first validate that the decomposition from equation 24 accurately tracks CES prices:

$$\Delta \log p_k = \beta_1 \text{Direct}_k + \beta_2 \text{Supplier}_k + \beta_3 \text{Own-cust}_k + \beta_4 \text{Supp.-cust}_k + \beta_5 \text{Income}_k + \epsilon_k. \quad (\text{F.2})$$

The β -coefficients measure how well each model channel predicts the equilibrium price when channels are exact (at $\sigma = 1$). Given the model structure, all coefficients should equal one by construction, with deviations indicating either numerical errors or CES-specific effects prevalent even with $\sigma = 1$.

Results: As shown in table F.1, coefficients obtained from regression F.2 are indeed very close to one, with a high $R^2 = 0.9997$. These results confirm that the decomposition of equation 24 exactly captures equilibrium price responses in the simulation, with minor deviations representing numerical rounding.

F.3 Regression 2: Empirical proxy regression

We now estimate the reduced-form proxy regression that mirrors the empirical specification in equation 8:

$$\Delta \log p_k = \gamma_1 (\text{st}\phi)_k + \gamma_2 (\text{st}[H\phi])_k + \gamma_3 (\text{st}[(Z - I)\phi])_k + \epsilon_k. \quad (\text{F.3})$$

This regression tests whether the empirical proxy system in section 4.4 recovers the model's structural parameters in finite samples with observationally relevant noise and collapsing dimensions.

The predicted signs from theory are: $\gamma_1 > 0$ (direct cost), $\gamma_2 > 0$ (upstream supplier cost), $\gamma_3 < 0$ (downstream customer demand). The magnitude of R^2 reflects how much heterogeneity in prices is explained by the three proxy channels.

Results: As shown in table F.1, the predicted signs are recovered in the results for regression model F.3, with $\hat{\gamma}_1 = 1.610 > 0$, $\hat{\gamma}_2 = 0.912 > 0$, and $\hat{\gamma}_3 = -2.163 < 0$.

The $R^2 = 0.5128$ indicates that these three channels explain approximately half of cross-sectional price variation, a strong result for a three-parameter parsimonious design. The fitted scatter plot (right panel of figure F.1) shows tight correspondence between predicted and actual prices.

Table F.1: Exact model vs. simulation-based regression results

Regression 1: Regression 2:

Channel decomposition Empirical proxies

Direct cost (+) 1.016

Supplier cost (+) 1.014

Own customers (−) 0.997

Suppliers' customer (−) 1.015

Income 1.300

$i \cdot \phi_k$ (direct, +) 0.339

$$i \cdot [(H - l)\phi]k \text{ (upstream, +)} 0.879$$

$$i \cdot [(Z - l)\phi]k \text{ (downstream, -)} -2.086$$

$$R^2 0.9990.482$$

Notes: Coefficients from decomposition regression on exact model channels (Regression 1), and simulation-based regression (Regression 2), following equation 28.

F.4 Channel sign analysis

To assess whether the model's sign predictions hold uniformly or only on average, we compute the share of observations where each channel has the predicted sign.

Interpretation: As reported in table F.2, the four non-ambiguous channels in equation 24 (direct, supplier, own-customers, suppliers' customers) display perfect sign recovery. The income channel shows mixed signs as it depends on the balance of cost-push and demand-pullback forces, which varies across sectors. In the empirical estimation, time fixed effects absorb this aggregate income term, leaving it unidentified.

F.5 Channel contribution diagnostics

This subsection discusses the relative quantitative contribution of each channel to explaining price variation. Figure F.2 and table F.3 summarize the channel-contribution decomposition used to benchmark the robustness exercises.

Key findings:

Figure F.1: Predicted versus simulated prices (Regressions 1 and 2)

Notes: Fitted model from decomposition regression on exact model channels (Regression 1, left panel), and simulation-based regression (Regression 2, right panel), following equation 28.

Figure F.2: Channel contributions: distribution and composition

Notes: Left panel: histogram of contributions by channel. Middle and right panels: stacked contributions showing which channels dominate price responses across the distribution.

1. Symmetric cost-demand split: Direct and supplier channels contribute approximately equally to price variation, as do own-customer and suppliers' customer channels. This reflects the structural balance between cost-push and demand-pullback forces.

2. Income channel smaller but structural: The income effect contributes less on average but plays a material role in reconciling the aggregate constraint across sectors. Its covariance share shows the expected negative correlation with other channels as when income rises, other channels contract to satisfy the numéraire condition.

Table F.2: Sign analysis by decomposition channel

Expected

Channelsign% correct% positiveMean

(i) Direct cost+100.0100.00.195

(ii) Supplier cost+100.0100.00.130

(iii) Own customers–100.00.0–0.312

(iv) Suppliers' cust.–100.00.0–0.209

(v) Incomen/an/a0.0–0.012

Notes: Share of sector-shock observations with correctly model-predicted signs (third column), unconditional share of positive sign (fourth column), mean channel contribution (fifth column).

3. Negative covariance shares: Covariance shares can be negative when a channel offsets price variation in other channels. This reflects the structural interdependence of channels in general equilibrium.

4. Consistency with empirical variance decomposition: As shown in table F.3, downstream customer channels account for the largest share of price variations when considering a fully model-consistent decomposition of the price-interest rate elasticity.

F.6Robustness: Sign patterns across elasticity values

To check that our results are not driven by the choice $\sigma = 1$, we verify that sign patterns remain stable as σ varies. We re-run the full simulation exercise and regressions 1 and 2 for elasticity values spanning $\sigma \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$ and report the coefficient estimates and model fit statistics.

Key findings

Table F.4 shows the decomposition channel coefficients (Regression 1) across elasticity values. The regression coefficients remain tightly clustered around their theoretical values (direct/supplier/supp.-cust. near +1, own-customers near –1) for all σ values, with $R^2 > 0.99$ throughout. This confirms that decomposition 24 is appropriate for actual CES price responses even when elasticities deviate substantially from unity.

Table F.3: Channel decomposition

MeanMeanCovariance

Channelcontribution|contribution| (%)share (%)

(i) Direct cost0.19874.438.4

(ii) Supplier cost0.13249.711.3

(iii) Own customers–0.310116.868.2

(iv) Suppliers' cust.–0.21279.7–18.0

(v) Income-0.0165.90.1

Total-0.208326.4100.0

Notes: Mean contribution in units of log prices (second column), mean absolute contribution share (explanatory power of each channel in $|p|$, third column), covariance share (share of channel's covariance in total price variance, fourth column). Covariance shares can be negative due to potentially offsetting channels.

Table F.5 reports the four-proxy empirical regression (Regression 2). Importantly, the predicted signs – positive for direct and upstream channels, and negative for downstream proxies – hold uniformly across the entire range of elasticities. While magnitudes vary – e.g., the downstream coefficient ranges from -1.6 to -2.8 as σ increases – , the sign patterns persist across the range of values for σ . The R^2 remains in the range 0.45 – 0.55 for all values, indicating stable explanatory power.

Table F.4: Robustness: Decomposition coefficients across elasticity values

Channel $\sigma = 0.50\sigma = 0.75\sigma = 1.00\sigma = 1.25\sigma = 1.50\sigma = 2.00$

Direct cost 1.09841.04281.00000.96350.93150.8828

Supplier cost 1.08511.04281.00000.94060.94830.9219

Own customers 0.98010.99211.00001.00851.01341.0249

Suppliers' customers 1.09741.04481.00000.95290.94750.9251

Income 2.77241.78281.00000.2624-0.3953-1.3895

R^2 0.98840.99761.00000.99850.99460.9839

Obs. 105001050010500105001050010500

Notes: Regression 1 coefficients (five CD channels) estimated for each elasticity value σ . "Obs." is the number of sector observations.

Table F.5: Robustness: Proxy regression coefficients across elasticity values

Regressor $\sigma = 0.50\sigma = 0.75\sigma = 1.00\sigma = 1.25\sigma = 1.50\sigma = 2.00$

$s \times \phi_j$ 0.38040.35280.34540.32460.29540.2879

$s \times (H - I) * \phi_j$ 0.90070.87520.86570.87440.84530.8284

$s \times (Z - I) * \phi_j$ -2.1951-2.1189-2.0843-2.0376-1.9748-1.9331

R^2 0.44250.42780.40770.40750.37280.3635

Obs. 105001050010500105001050010500

Notes: Regression 2 coefficients (three empirical proxies) estimated for each elasticity value σ .

All three predicted signs (direct: +, upstream: +, downstream: -) hold across every elasticity.

GA two-period flexible-price extension

The benchmark model in section 4 is static and uses the consumption price index as the numéraire. This appendix shows that the sectoral price decomposition derived

there is not an artifact of this normalization. We extend the environment to two periods while keeping prices fully flexible. The only dynamic force we add is the household Euler equation, which pins down current nominal expenditure relative to future nominal expenditure. As a result, a monetary tightening lowers the current aggregate price level through a common nominal scale effect. At the sector level, however, prices continue to move according to the same direct, upstream, and downstream financial-friction channels, plus this common aggregate component.

G.1 The two-period economy

The economy lasts for two periods, $t = 0, 1$. Monetary policy affects sector j 's effective variable input costs through the same working-capital wedge used in the benchmark model:

$$q_{jt} = 1 + it\phi_j, \phi_j \geq 0. \quad (G.1)$$

Production is otherwise identical to the static environment. Sector j operates a constant-returns Cobb–Douglas technology,

$$\omega_{kj}k_{j,t}l_{jt} = \bar{l}_j, \quad (G.2)$$

$$a_{j,t}$$

$$Y$$

$$y_{jt} = a_{jl}$$

$$x$$

$$k$$

with constant returns to scale,

$$X$$

$$\alpha_j +$$

$$\omega_{kj} = 1. \quad (G.3)$$

$$k$$

As in the main text, ω_{kj} denotes the share of sector j 's costs spent on inputs from sector k , so columns of Ω correspond to customer sectors. The vector of labor shares therefore satisfies

$$\alpha = (I - \Omega')^{-1} \mathbf{1}. \quad (G.4)$$

The corresponding cost-based Leontief inverse is

$$H = (I - \Omega')^{-1}. \quad (G.5)$$

The Cobb–Douglas adding-up condition implies the same identity used throughout the benchmark derivation:

$$H\alpha = \mathbf{1}. \quad (G.6)$$

Finally, define

$$D\alpha = \text{diag}(\alpha). \quad (G.7)$$

G.2 Firm optimality and prices

Let

$$R_{jt} = p_{jt} y_{jt} \quad (G.8)$$

denote nominal revenue in sector j and period t . Cost minimization implies that the financing wedge scales both labor and intermediate-input costs:

$$q_{jt} w_{jt}^{-1} l_j = \alpha_j R_{jt}, \quad (G.9)$$

and

$$q_{jt} p_{kt} x_{kj,t} = \omega_{kj} R_{jt}. \quad (G.10)$$

Equivalently,

$$w_{jt} = \alpha_j R_{jt}$$

$$, \quad (G.11)$$

$$q_{jt}^{-1} l_j$$

and

$$p_{kt} x_{kj,t} = \omega_{kj} R_{jt}$$

$$. \quad (G.12)$$

$$q_{jt}$$

Substituting these factor demands into the unit-cost condition yields a log-price equation that is the dynamic counterpart of equation 20 in the main text:

X

$$\log p_{jt} = \chi_j + (1 - \alpha_j) \log q_{jt} + \alpha_j \log R_{jt} +$$

$$\omega_{kj} \log p_{kt}, \quad (G.13)$$

k

$$\text{where } \chi_j = -\log \alpha_j - \alpha_j \log^{-1} l_j - \alpha_j \log \alpha_j - P$$

k $\omega_{kj} \log \omega_{kj}$ collects sector-specific

constants. Stacking sectors and inverting the input-output block gives

$$\log p_t = H [\chi + (I - D\alpha) \log q_t + D\alpha \log R_t] \quad (G.14)$$

G.3 Revenue system

Households allocate final expenditure across sectoral goods with Cobb–Douglas shares v :

X

$$D_{jt} = v_j E_t, E_t = P^c$$

$$t \ C_t,$$

$$v_j = 1. \quad (G.15)$$

j

Sectoral revenue equals final demand plus sales to all downstream producers,

X

$$R_{jt} = v_j E_t +$$

$$p_{jtxjm,t} \cdot (G.16)$$

m

Using the input demand of customer sector m ,

$$p_{jtxjm,t} = \omega_{jm} R_{mt}$$

$$\cdot (G.17)$$

q_{mt}

Thus the wedge affects the revenue system by reducing the revenue share paid

to intermediate-input suppliers:

R_{mt}

X

$$R_{jt} = v_j E_t +$$

ω_{jm}

$$\cdot (G.18)$$

q_{mt}

m

In vector form,

$$R_t = E_t v + \Omega Q^{-1}$$

$$t R_t, (G.19)$$

where

$$Q_t = \text{diag}(q_{1t}, \dots, q_{Nt}). (G.20)$$

Solving for revenues gives

$$-1 \ v. (G.21)$$

$$I - \Omega Q^{-1}$$

$$R_t = E_t$$

t

It is useful to collect the dependence of revenue shares on the wedge vector in

the vector

$$-1 \ v. (G.22)$$

$$I - \Omega Q^{-1}$$

$$z(q_t) \equiv$$

t

Then sectoral revenues can be written as

$$R_t = E_t z(q_t), (G.23)$$

and therefore

$$\log R_t = \log E_t + \log z(q_t). (G.24)$$

G.4 Nominal scale and relative prices

The central object in this appendix is the separation between aggregate nominal scale and sectoral relative prices. Substituting equation G.24 into the price system G.14 gives

$$\log p_t = H [\chi + (1 - \alpha) \log q_t + \alpha \log z(q_t) + \alpha \log E_t]. (G.25)$$

Using the identity $H\alpha = 1$, this becomes

$$\log p_t = \log E_t + e_p(q_t), (G.26)$$

where

$$e_p(q_t) \equiv H [\chi + (1 - \alpha) \log q_t + \alpha \log z(q_t)]. (G.27)$$

Equation G.26 is the key link to the benchmark decomposition. Nominal expenditure E_t shifts all sectoral prices one-for-one, while the vector $e_p(q_t)$ captures the relative-price effects generated by financial wedges and production-network linkages.

G.5 Aggregate consumption

The Cobb-Douglas consumption price index satisfies, up to constants,

$$t = v' \log p_t. (G.28)$$

$$\log P_c$$

Using the decomposition in equation G.26,

$$t = \log E_t + v' e_p(q_t). (G.29)$$

$$\log P_c$$

Since $E_t = P_c$

$$t - C_t,$$

$$\log C_t = \log E_t - \log P_c$$

$$t. (G.30)$$

Therefore,

$$\log C_t = -v' e_p(q_t). (G.31)$$

Real consumption is therefore pinned down by the relative price distortion induced by q_t . The aggregate nominal scale E_t affects nominal prices and nominal expenditure, but not feasible real consumption directly.

G.6 Household Euler equation

The intertemporal margin is introduced through the household Euler equation,

$$= \beta(1 + i_0) P_c$$

$1C_0$

$0P_c$

.(G.32)

1

Using P_c

$t = E_t/C_t$, this condition can be rewritten as

$1C_0$

$= \beta(1 + i_0)E_0/C_0$

$1C_1$

.(G.33)

E_1/C_1

The real-consumption terms cancel. Hence the Euler equation determines the relative nominal expenditure levels across the two periods:

$1 = \beta(1 + i_0)E_0E_1$

.(G.34)

Hence

E_0E_1

$= 1/\beta(1 + i_0)$.(G.35)

Equivalently,

$\log E_0 = \log E_1 - \log \beta - \log(1 + i_0)$.(G.36)

This is the source of monetary non-neutrality in the two-period flexible-price extension. A higher current interest rate lowers current nominal expenditure relative to future nominal expenditure. Through equation G.26, this generates a common decline in current sectoral prices, while real consumption continues to be governed by the relative-price distortion in equation G.31.

G.7 Exact response to an interest-rate change

Consider a change in the period-0 interest rate from i to i' , holding future nominal expenditure E_1 and the future policy rate i_1 fixed. Since

$q_j(i) = 1 + i\phi_j$.(G.37)

the Euler equation implies

$\Delta \log E_0 = -\log(1 + i')$

$1 + i$.(G.38)

Real consumption changes by

$\Delta \log C_0 = -v' [ep(q(i')) - ep(q(i))]$.(G.39)

The consumption price index changes according to the accounting identity P_c

$$0 =$$

$$E_0/C_0:$$

$$\Delta \log P_c$$

$$0 = \Delta \log E_0 - \Delta \log C_0. (G.40)$$

Therefore,

$$0 = -\log 1 + i'$$

$$1 + i + v' [\epsilon_p(q(i')) - \epsilon_p(q(i))] . (G.41)$$

$$\Delta \log P_c$$

Similarly, sectoral prices satisfy

$$\Delta \log p_0 = -\log 1 + i'$$

$$1 + i + v' [\epsilon_p(q(i')) - \epsilon_p(q(i))] . (G.42)$$

Equations G.41 and G.42 show that a monetary tightening has two effects. The first is a common nominal scale effect, which moves all sectoral prices in the same direction. The second is a relative price effect, which is generated by the same network-induced financial wedges that appear in the static decomposition.

G.8 Sectoral quantity responses

The decomposition also clarifies how sectoral quantities adjust. Sectoral revenues satisfy

$$R_0 = E_0 z(q_0). (G.43)$$

Hence

$$\Delta \log R_0 = \Delta \log E_0 + \Delta \log z(q_0). (G.44)$$

Sectoral household consumption is

$$c_{j0} = v_j E_0$$

$$. (G.45)$$

$$p_{j0}$$

The common nominal scale effect cancels between E_0 and p_{j0} , so

$$\Delta \log c_0 = -\Delta \epsilon_p(q_0). (G.46)$$

Sectoral output is

$$y_{j0} = R_{j0}$$

$$. (G.47)$$

$$p_{j0}$$

Thus its response combines the revenue-share adjustment and the relative price adjustment:

$$\Delta \log y_0 = \Delta \log z(q_0) - \Delta \epsilon_p(q_0). (G.48)$$

Intermediate inputs satisfy

$$x_{kj,0} = \omega_{kj} R_{j0}$$

, (G.49)

$$q_{j0} p_{k0}$$

so their response reflects the customer's revenue response, the customer's financing wedge, and the supplier's price:

$$\Delta \log x_{kj,0} = \Delta \log R_{j0} - \Delta \log q_{j0} - \Delta \log p_{k0}. \quad (G.50)$$

Finally, wages satisfy

$$w_{j0} = \alpha_j R_{j0}$$

. (G.51)

$$q_{j0}^{-1} l_j$$

Hence

$$\Delta \log w_{j0} = \Delta \log R_{j0} - \Delta \log q_{j0}. \quad (G.52)$$

G.9 Local response around an arbitrary interest rate

We now differentiate the exact expressions around an arbitrary initial interest rate

i. The relevant semi-elasticity of the financing wedge with respect to $\log(1 + i)$ is

$$s_j(i) \equiv \partial \log q_j(i)$$

$$\partial \log(1 + i) = (1 + i) \phi_j$$

$$1 + i \phi_j, \quad (G.53)$$

and we collect these terms in

$$s(i) = (s_1(i), \dots, s_N(i))'. \quad (G.54)$$

The revenue share objects evaluated at the initial interest rate are

$$I - \Omega Q(i)^{-1} - 1, \quad (G.55)$$

$$Z(i) =$$

$$z(i) = Z(i) v, \quad (G.56)$$

and

$$B(i) = \text{diag}(z(i))^{-1} Z(i) \Omega Q(i)^{-1} \text{diag}(z(i)). \quad (G.57)$$

Differentiating the revenue-allocation vector gives

$$d \log z(i) = -B(i) d \log q(i). \quad (G.58)$$

Combining this derivative with the relative price object in equation G.27 yields

$$\text{dep}(i) = H [(I - D\alpha) - D\alpha B(i)] d \log q(i). \quad (G.59)$$

Since

$$d \log q(i) = s(i) d \log(1 + i), \quad (G.60)$$

the relative price response is

$$\text{dep}(i) = H [(I - D\alpha) - D\alpha B(i)] s(i) d \log(1 + i). \quad (G.61)$$

The consumption-weighted relative price response is summarized by

$$\kappa(i) \equiv v'H [(I - D\alpha) - D\alpha B(i)] s(i). (G.62)$$

Then aggregate real consumption and the aggregate price index satisfy

$$d \log C_0 = -\kappa(i) d \log(1 + i_0), (G.63)$$

and

$$d \log P_c$$

$$0 = [\kappa(i) - 1] d \log(1 + i_0). (G.64)$$

Sectoral prices respond according to

$$d \log p_0 = -d \log(1 + i_0)^{1+H} [(I - D\alpha) - D\alpha B(i)] s(i) d \log(1 + i_0). (G.65)$$

The first term is the common nominal-scale effect implied by the Euler equation.

The second term is the sector-specific relative-price component. It has the same economic content as the benchmark decomposition: direct financing costs, upstream cost propagation, and downstream revenue compression.

G.10 The zero-interest case

The comparison with the benchmark model is especially transparent around a zero-interest baseline. Set

$$i = 0. (G.66)$$

Then

$$q_j(0) = 1, Q(0) = I, (G.67)$$

and

$$s_j(0) = \phi_j. (G.68)$$

Moreover,

$$Z(0) = (I - \Omega)^{-1} \equiv Z, (G.69)$$

and

$$z(0) = Zv. (G.70)$$

The revenue-derivative matrix becomes

$$B_0 = \text{diag}(z) - Z\Omega \text{diag}(z). (G.71)$$

The relative-price derivative is

$$\text{dep}(0) = H [(I - D\alpha) - D\alpha B_0] \phi \, di_0. (G.72)$$

At $i = 0$, the undistorted equilibrium satisfies the envelope condition

$$v'H [(I - D\alpha) - D\alpha B_0] = 0'. (G.73)$$

Therefore,

$$\kappa(0) = 0. (G.74)$$

It follows that

$$d \log C_0 = 0. (G.75)$$

A small monetary tightening around a zero-interest baseline has no first-order

effect on feasible aggregate consumption.

Since $d \log(1 + i_0) = di_0$ at $i_0 = 0$, the aggregate price index satisfies

$$d \log P_c$$

$$0 = -di_0. (G.76)$$

Sectoral prices satisfy

$$d \log p_0 = -di_0 \frac{1}{1 + H} [(1 - \alpha) - \alpha B_0] \phi di_0. (G.77)$$

The second term changes relative prices across sectors, but its consumption-weighted aggregate is zero:

$$v'_{dep}(0) = 0. (G.78)$$

Thus the aggregate price-index response is entirely given by the common nominal scale effect:

$$d \log P_c$$

$$0 = -di_0. (G.79)$$

The sectoral deviations around this aggregate decline are governed by the same network-induced financial forces emphasized in the main model. The dynamic extension therefore changes the aggregate normalization—a tightening lowers the current price level—but leaves the sectoral decomposition intact.

Acknowledgements We thank Hassan Afrouzi, Christian Bayer, Benjamin Born, Vasco Carvalho, Lawrence Christiano, Rubén Domínguez-Díaz, Mishel Ghassibe, Òscar Jordà, Joachim Jungherr, Christophe Kamps, Sujit Kapadia, Keith Kuester, Philip R. Lane, Javier Quintana, Farzad Saidi, Donghai Zhang and participants at the Banque de France Conference on “Financial Intermediation and Monetary Policy: Recent Trends and New Challenges”, the 2025 ECB Inflation Drivers and Dynamics Conference, the 28th Conference “Theories and Methods in Macroeconomics”, the 52nd OeNB-SUERF Annual Economic Conference, the 7th ChaMP Research Network Workshop and the 40th Annual Congress of the European Economic Association for valuable comments. Any remaining errors are our own. The views in this paper are those of the authors and do not necessarily reflect the views of the European Central Bank or the Eurosystem.

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PDF ISBN 978-92-899-7995-5 ISSN 1725-2806 doi: 10.2866/7077392 QB-01-26-199-EN-N

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